

Chronic inflammation healing and repair

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Objectives

- 1-Definition and causes of chronic inflammation
- 2- Identify the cells of chronic inflammation
- 3- Understand the etiology and main features of granulomatous inflammation
- 4- Understand the mechanism of tissue healing (regeneration and repair).
- 5- Recognize the sequence of events in tissue repair.
- 6- Recognize the differences between primary and secondary intention healing.
- 7- Factors that could affect healing process. 8- Understand the healing process of bone fracture.

Chronic Inflammation

- **Definition:**

- Inflammation of prolonged duration in which active inflammation, tissue injury and the healing proceed simultaneously.
- Occurs when acute response can not be resolved ,either because of the *persistence of the injurious agent* or because of the *interference in the normal process of healing* .

- **Causes:**

1. Viral infection.

- intracellular infection require lymphocytes & macrophage to eradicate the infection

2. Persistent microbial Infections

- Ex. *Tubercle bacilli* .
- Organism of low toxicity and evoke an immune reaction = **delayed hypersensitivity**

3. Prolonged Exposure to toxic Agents,

- Exogenous (*Silicosis*)
- Endogenous (Atherosclerosis)

4. Autoimmunity

- Ex. Autoimmune diseases (rheumatoid arthritis).

Chronic Inflammation

- **Morphologic Features:**

- **Infiltration with mononuclear cells** (macrophages, lymphocytes & plasma cells)

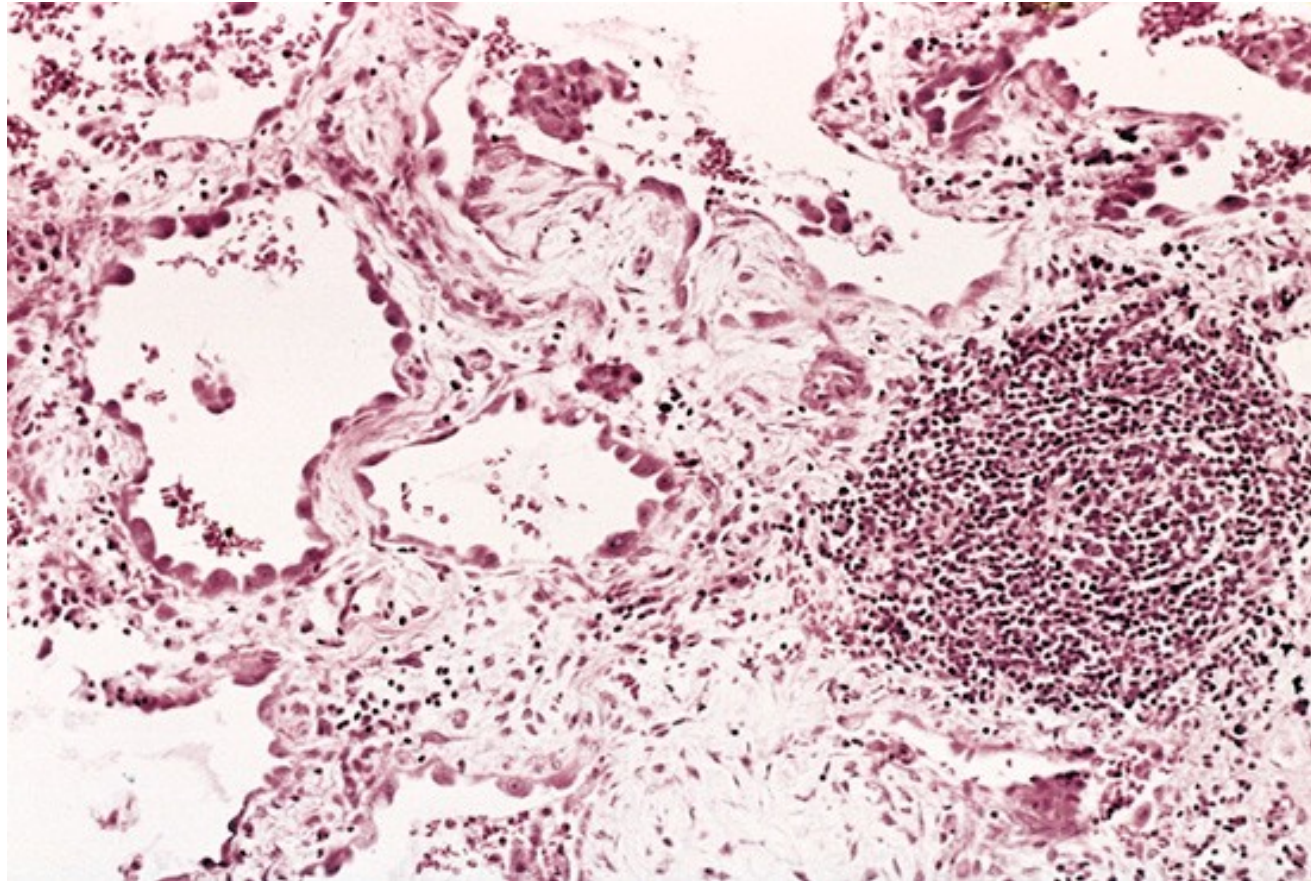
- indicates persistent reaction to injury

- **Tissue destruction**

induced by the persistent offending agent or by the inflammatory cells.

- **Repair involving angiogenesis and fibrosis**

- Attempt to replace lost tissue



Chronic inflammation in the lung showing all 3 characteristic histologic features:

- Collection of chronic inflammatory cells.***
- Destruction of parychyma.***
- Replacement by connective tissue (fibrosis)***

Cells of Chronic Inflammation

➤ Mononuclear Phagocyte System

- Circulating blood monocytes → Tissue macrophages

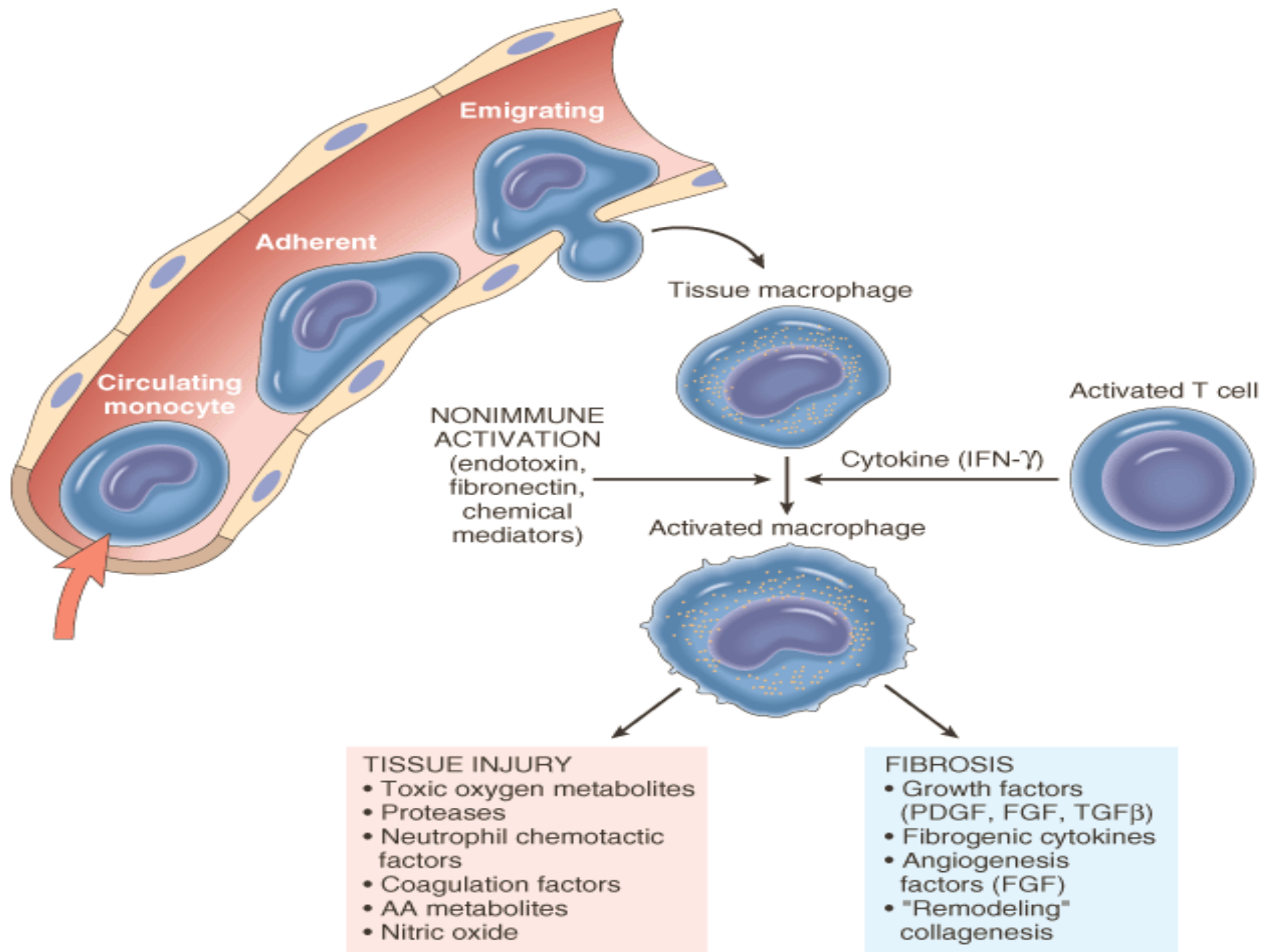


Kupffer cells (liver)

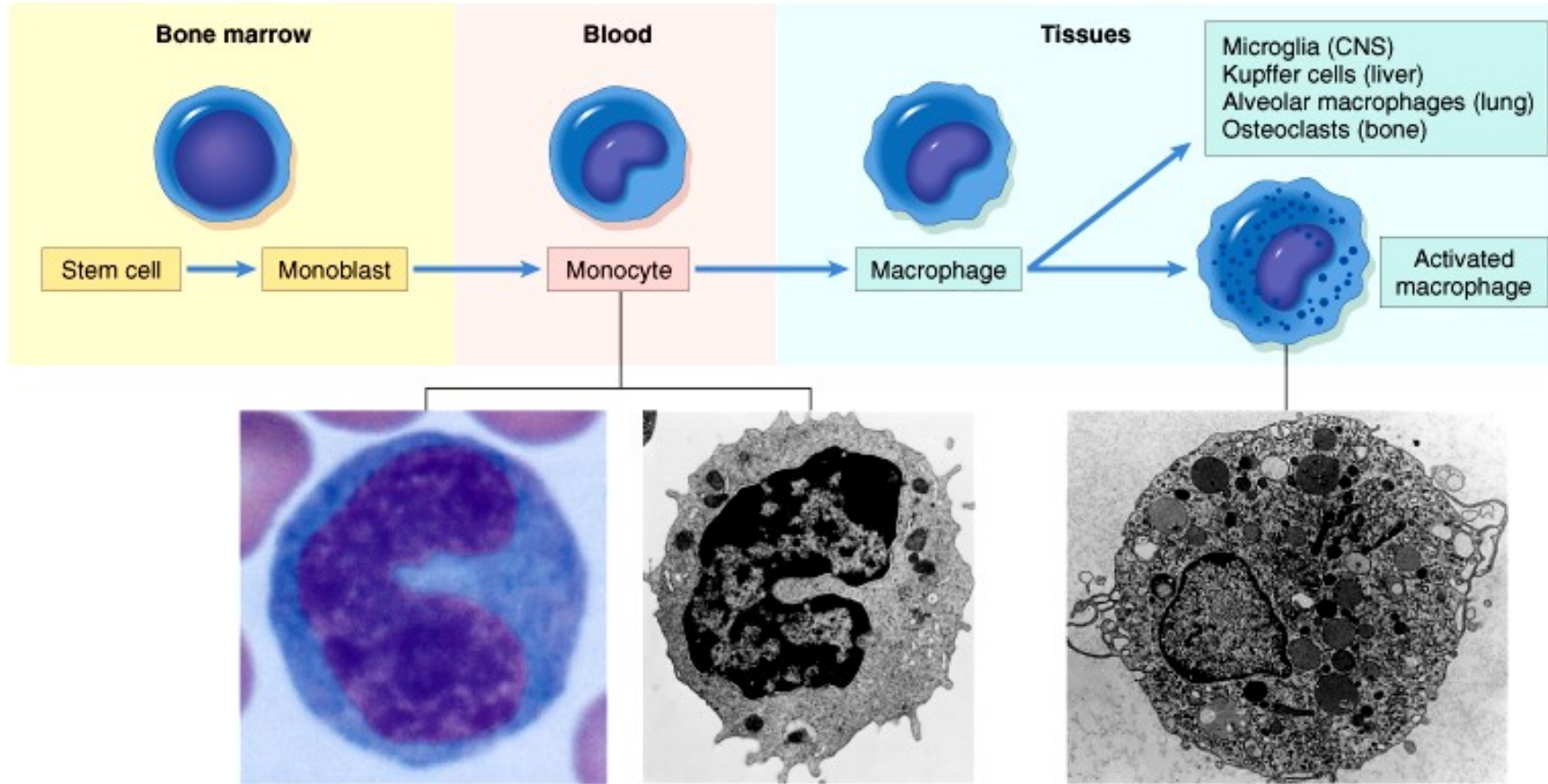
Sinus Histiocytes (spleen)

Microglia (CNS)

Alveolar Macrophages (lung)

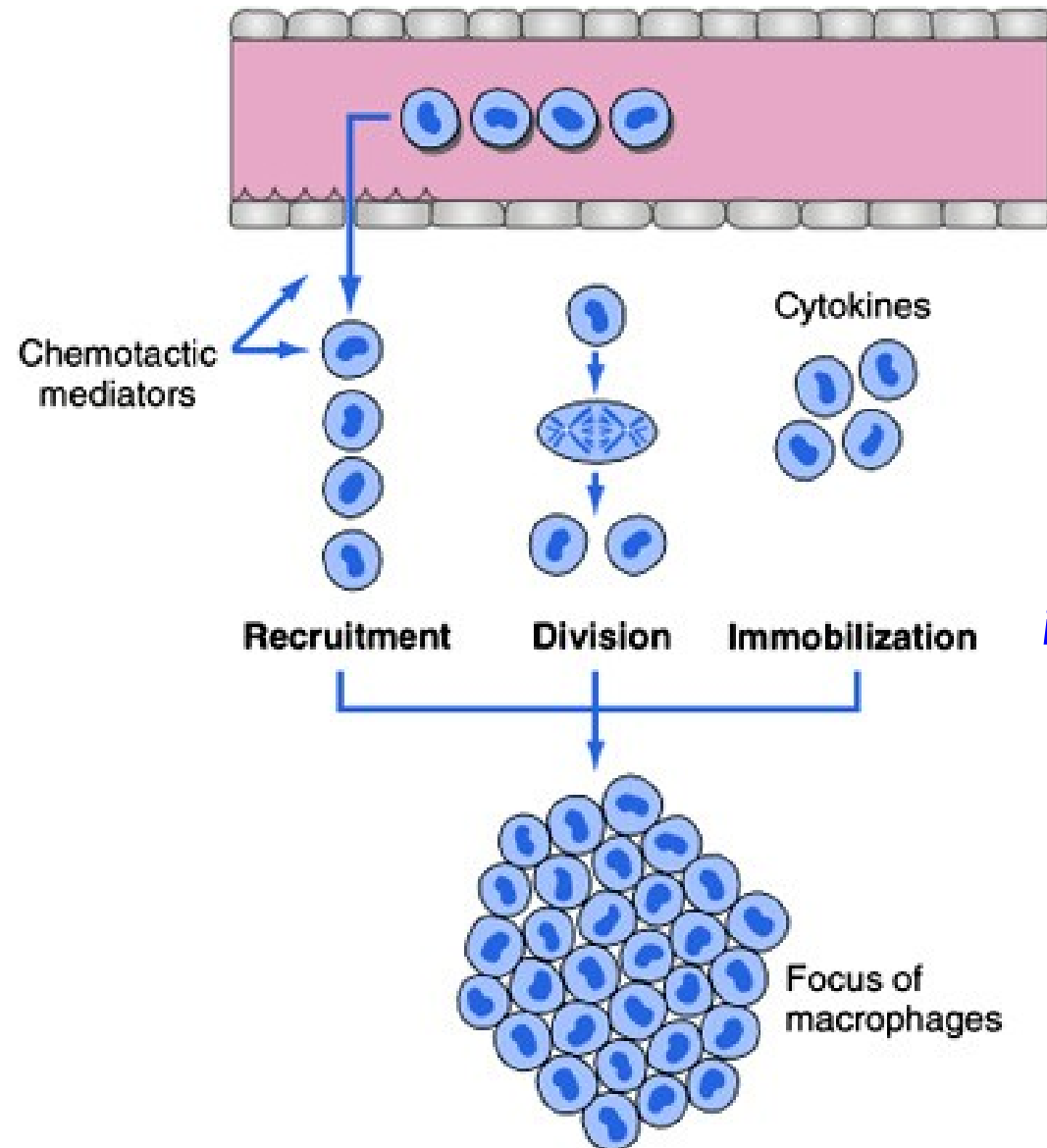


Maturation of Mononuclear Phagocytes



Mechanisms of macrophage accumulation during Chronic Inflammation

- Continued recruitment of monocytes from the circulation.
 - Most important source for macrophages
- Local proliferation of macrophages from the blood stream.
- Immobilization of macrophages within the site of inflammation
 - Cytokines and oxidized lipids can cause immobilization



Mechanisms of macrophage accumulation in tissues (the most important is continued recruitment from the microcirculation).

Other Cells in Chronic Inflammation:

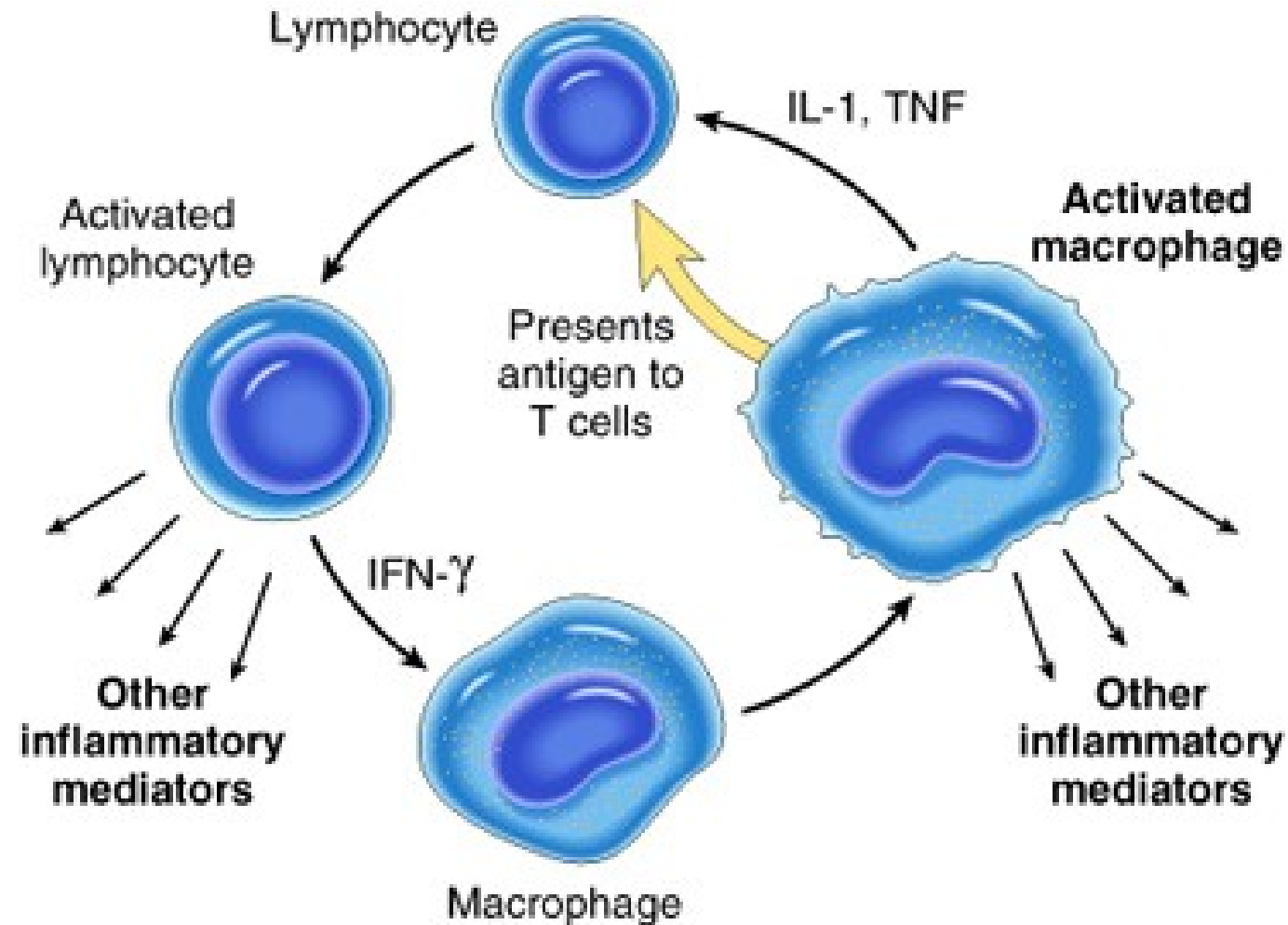
Other cell types present in chronic inflammation include lymphocytes, plasma cells, eosinophils, and mast cells.

➤ **Lymphocytes** are mobilized in both antibody-mediated & cell-mediated immune reactions and even in non-immune inflammation.

Lymphocytes & macrophages interact in a bidirectional way, and these reactions play an important role in chronic inflammation.

➤ **Eosinophils** are abundant in immune reactions mediated by IgE and in parasitic infections.

➤ **Mast cells** are widely distributed in connective tissues and participate in both acute and persistent inflammatory reactions.



Macrophage-lymphocyte interactions in chronic inflammation

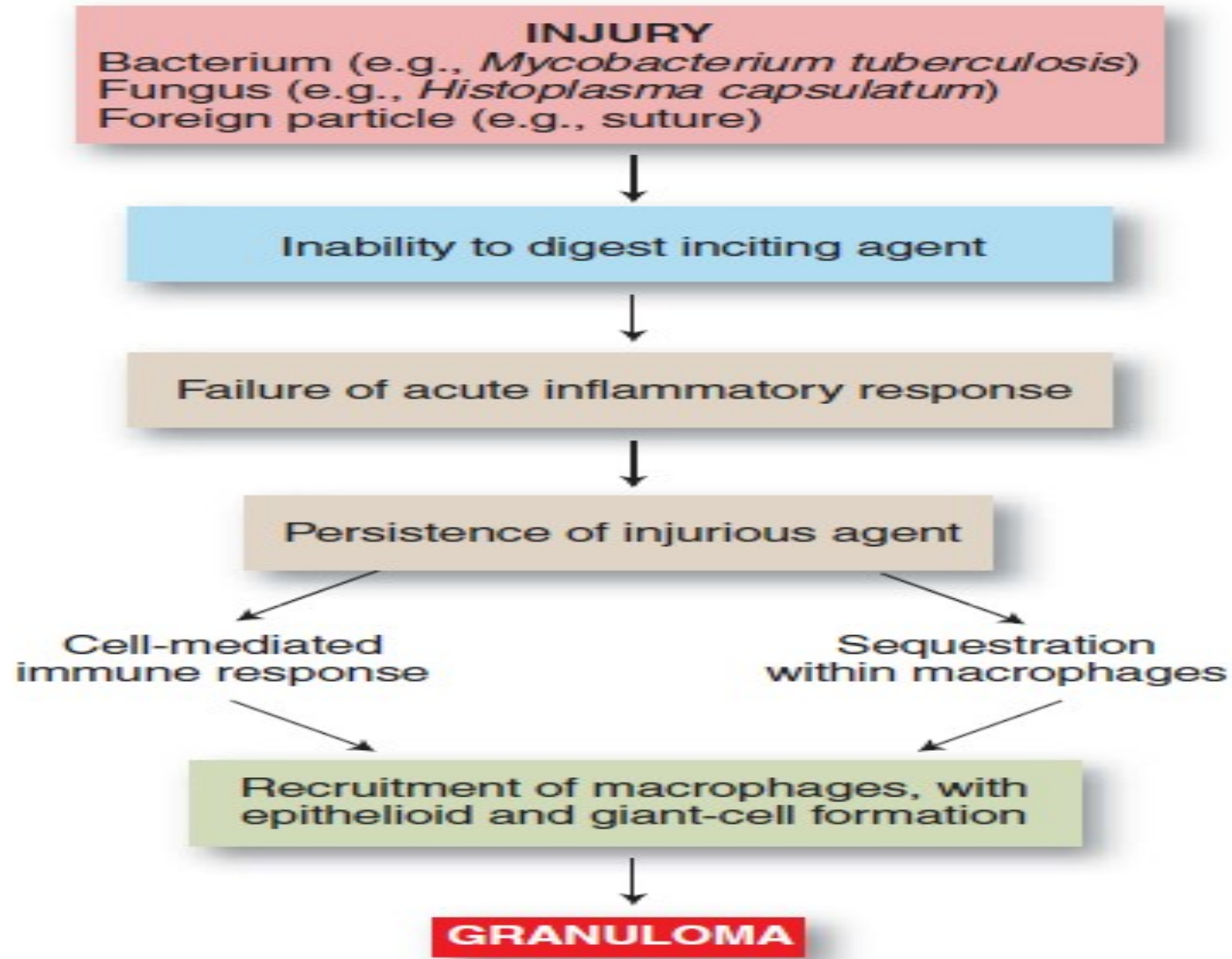
(macrophages and activated lymphocytes influence each other and also release inflammatory mediators that affect other cells)

Granulomatous inflammation

- *Granulomatous inflammation is a distinctive pattern of chronic inflammatory reaction characterized by focal accumulations of activated macrophages, which often develop an epithelial-like (epithelioid) appearance.*
- **GRANULOMA** - Nodular collection of **Epithelioid macrophages** surrounded by a rim of **lymphocytes & plasma cell**.

Granulomatous Inflammation

A distinctive pattern of chronic inflammation characterized by aggregation of activated macrophages that have been modified but acquiring an enlarged squamous cell-like appearance ,these cell called epithelioid cells

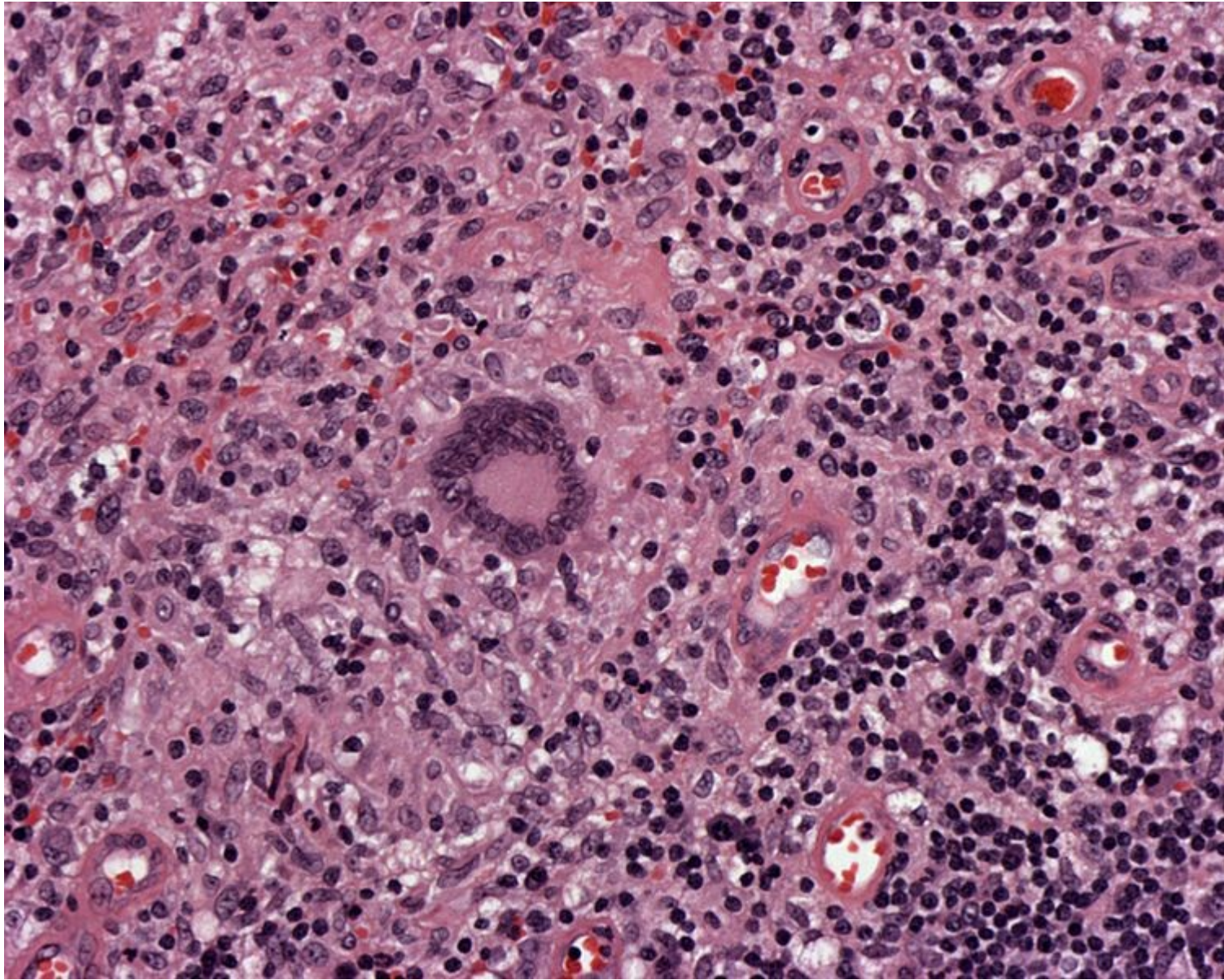


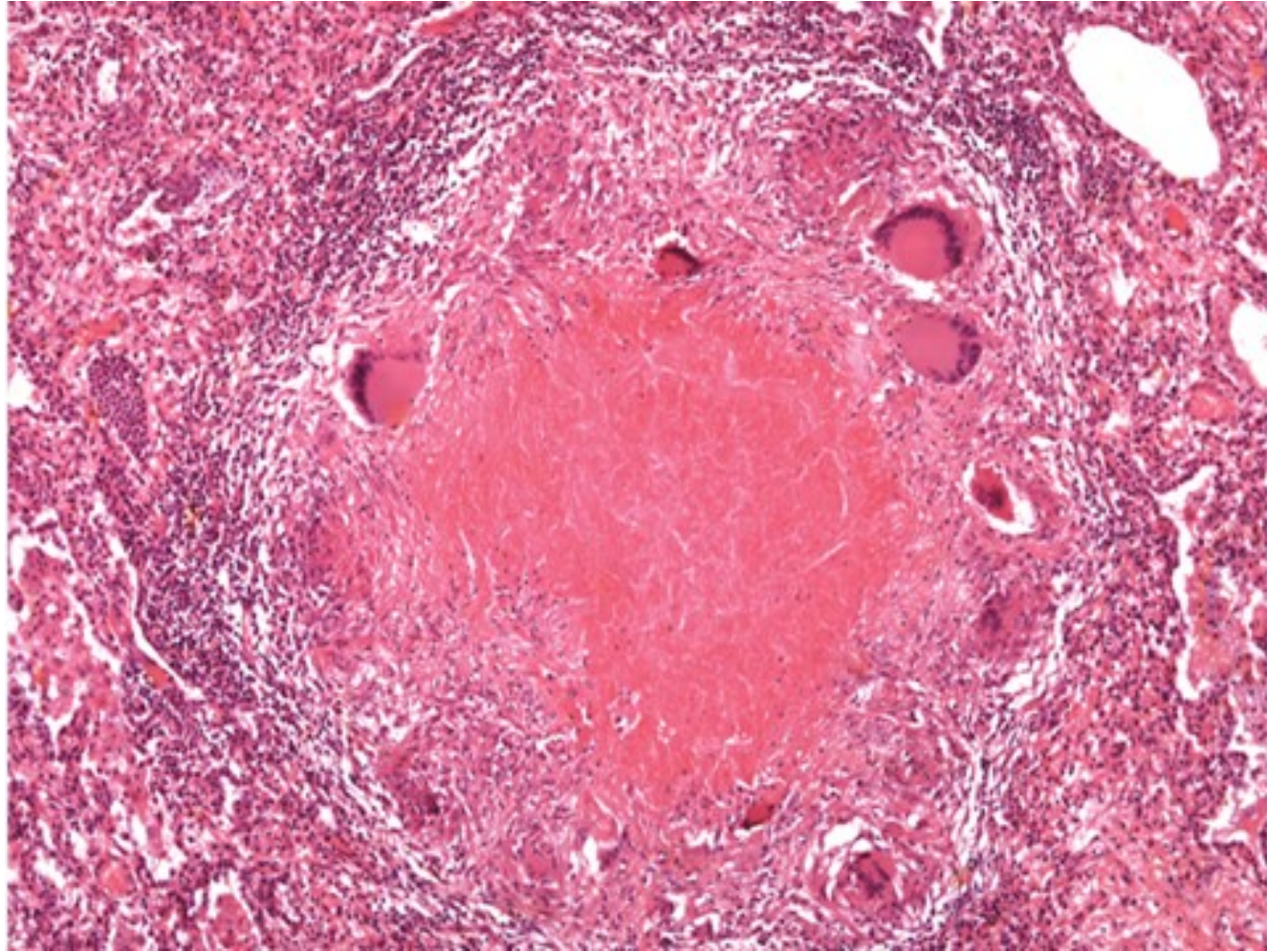
Morphology of **Granulomatous inflammation**

- In an H&E stain you can see:
 - Epithelioid cells have pale granular cytoplasm with indistinct boundaries
- Giant cells = Epithelioid cells that fuse (**Langhans**)
 - Can be found in the periphery or sometimes in the center of the granuloma
 - Have large mass of cytoplasm
 - Have 20 or more small nuclei arranged in the periphery (horseshoe-shaped) or haphazardly
- **Tuberculosis is the prototype of the granulomatous diseases**

Morphology of Granulomatous Inflammation:

- ❖ In the usual H & E preparations, activated macrophages (*epitheloid cells*) have *pink, granular cytoplasm with indistinct cell boundaries*.
- ❖ The aggregates of epitheloid macrophages are surrounded by *a collar of lymphocytes* secreting cytokines responsible for ongoing macrophage activation.
- ❖ Older granulomas develop a surrounding rim of *fibroblast & connective tissue (rim of scarring that usually contains the injurious agent)*.
- ❖ Frequently, *multinucleated giant cells* 40 - 50µm in diameter are also seen. They consist of a large mass of cytoplasm and multiple nuclei and they are derived from fusion of 20 or more macrophages.
- ❖ In granuloma associated with certain infectious organisms (TB), a combination of hypoxia & free radical injury leads to *a central zone of necrosis*.
- ❖ Grossly, this zone of necrosis has agranular cheesy appearance and is therefore called *caseous necrosis*.





Granulomatous inflammation

2 Types of Granulomatous Inflammation

- 1. **Immune** granulomas

- Caused by insoluble particles that are capable of inducing a cell-mediated immune response

Macrophages are transformed into Epitheloid cells and multinucleate giant cells

- Examples:

- Bacteria
 - Tuberculosis
 - Leprosy
- Parasites
 - Schistosomiasis (3 types)
- Fungi
 - Histoplasmosis
 - Blastomycosis

- sarcoidosis

Granulomatous inflammation

2.Foreign Body Granulomas

They are formed when the material is inert, too large and cannot be easily degraded by single macrophages (resists lysosomal degradation), and don't incite either an inflammatory or immune response.

Epithelioid cells and giant cells are apposed to the surface and encompass the foreign body.

The foreign body is usually found in the center of the granuloma.

Examples:

Metal/Dust

Silicosis

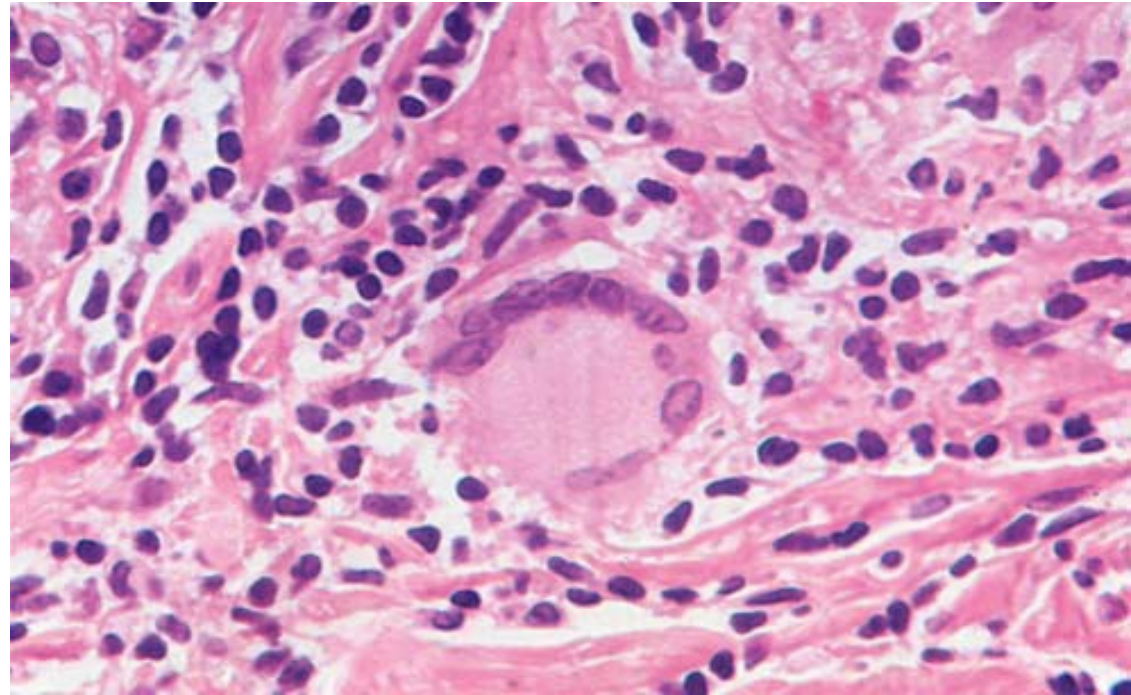
Foreign body

Splinter

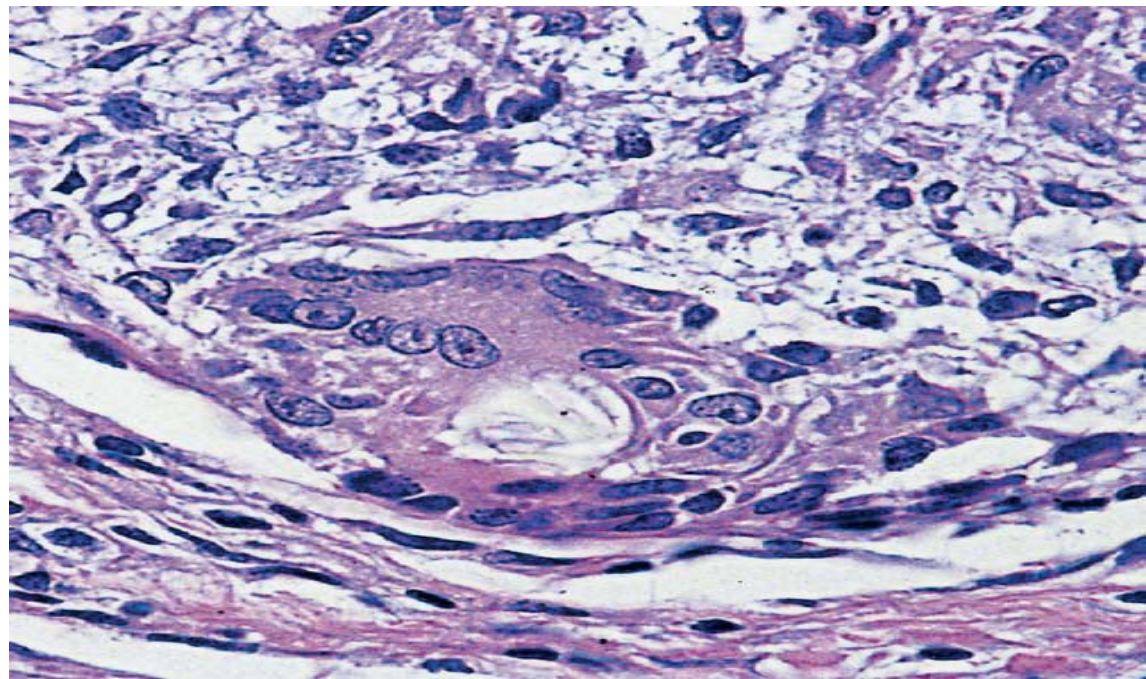
Suture

2 pattern of multinucleated giant cell

1.Langhans giant cell

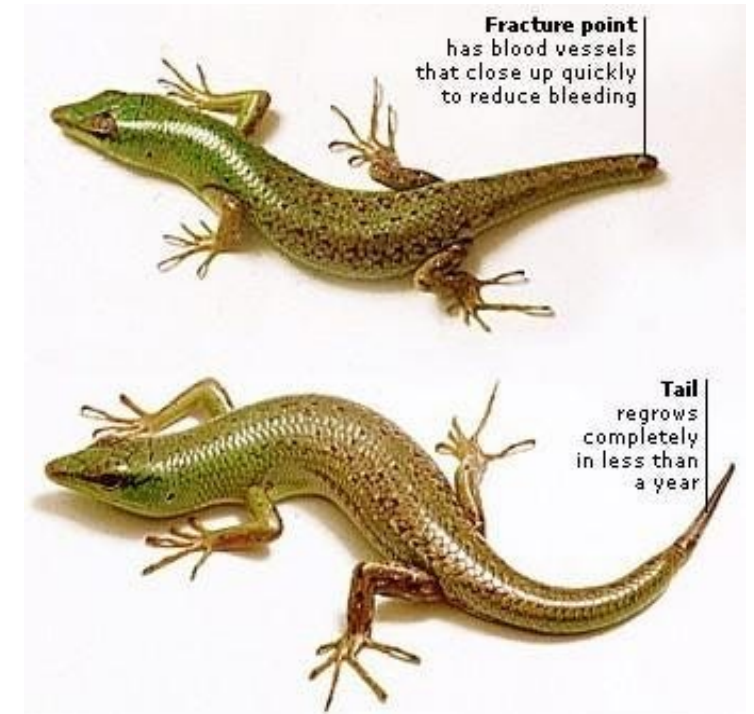


2.foreign body giant cell



REPAIR AND HEALING

- **Restoration of tissue architecture and function after an injury.**
- **Repair of damaged tissues occurs by one of two processes:**
 1. **Regeneration:** which is the restoration of normal cells.
 2. **Scar formation:** which is the deposition of connective tissue.



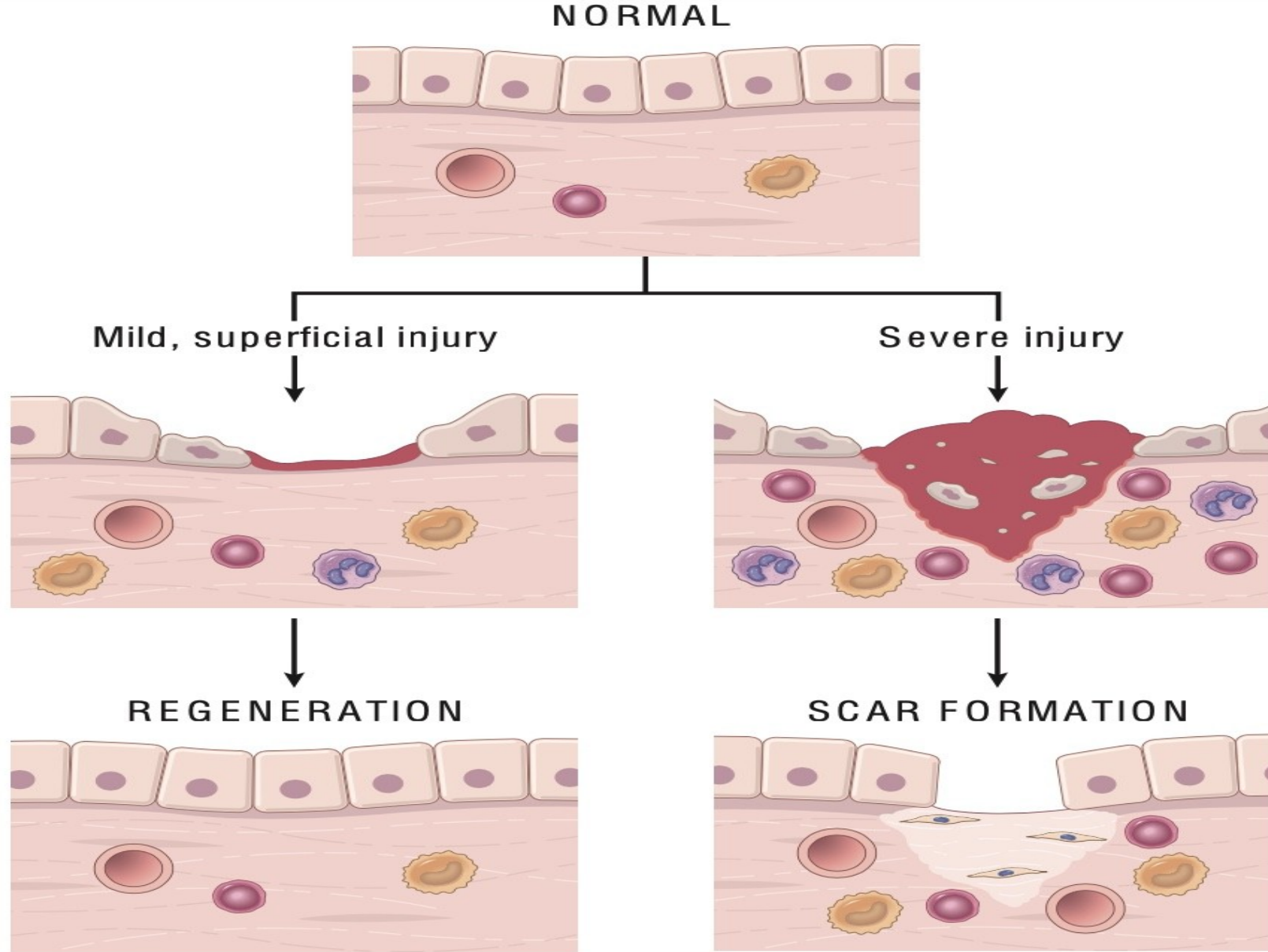


Figure 3.24 Mechanisms of tissue repair: regeneration and scar formation. Following mild injury, which damages the epithelium but not the underlying tissue, resolution occurs by regeneration, but after more severe injury with damage to the connective tissue, repair is by scar formation.

Tissue Proliferative Activity

- Tissues are divided into three groups according to their proliferative capacity:

Labile tissues. (*continuously dividing*) Cells of these tissues are continuously being lost and replaced. Labile cells include

- a)hematopoietic cells in the bone marrow
- b)the majority of surface epithelia, such as the stratified squamous epithelia of the skin, oral cavity

Stable tissues. Cells of these tissues are quiescent (in the G0 stage of the cell cycle) and have only minimal proliferative activity in their normal state. However, these cells are capable of dividing in response to injury or loss of tissue mass. Stable cells include:

- a)arenchymal cells of most solid tissues, such as liver, kidney, and pancreas.

Permanent tissues. The cells of these tissues are considered to be terminally differentiated and nonproliferative in postnatal life. In permanent tissues, repair is typically dominated by scar formation. Permanent cells include:

- a)neurons
- b)cardiac muscle

- So, depending on the proliferative activity of tissue, in addition to other criteria the type of tissue repair will be determined.**

REGENERATION

- Restoration of damaged tissues. This happens when:
 1. the tissue is able to replace the damaged components
 2. the residual tissue is structurally intact.
- Cell proliferation is controlled by growth factors and interactions of cells with the ECM.
- Regeneration accomplished by the help of two types of cells:
 1. *Differentiated cells* that survive the injury ex. Hepatocytes in the liver.
 2. *Tissue stem cells*, particularly the epithelia of the skin and intestines.
- Best example on tissue regeneration is the liver.

SCAR FORMATION (*Connective tissue deposition*)

- **Replacement of the injured cells with connective (fibrous) tissue.** Scar tissue replacement occurs in the following conditions:
 1. If the injured tissues are incapable of regeneration at all
 2. If repair cannot be accomplished by regeneration alone
 3. If the supporting structures of the tissue are too severely damaged to support regeneration of the tissue cells.



Steps in Scar Formation

- Repair by connective tissue deposition consists of sequential processes that follow tissue injury.

1. Hemostatic plug: Within minutes after injury, hemostatic plug composed of platelets is formed, which stops bleeding, provides a scaffold for the deposition of fibrin and release chemokines and inflammatory mediators.
2. Inflammation. Within 6-48 hours release of mediators with inflammatory cells recruitment to eliminate the offending agents, such as microbes that may have entered through the wound, and clear the debris. As the injurious agents and necrotic cells are cleared, the inflammation resolves.
3. Cell proliferation. In the next stage, which takes up to 10 days, several cell types, including epithelial cells, endothelial cells, and fibroblasts, proliferate and migrate to close the now clean wound.
 - a. Epithelial cells migrate over the wound to cover it up
 - b. Endothelial cells and pericytes proliferate to form new blood vessels, a process known as *angiogenesis*
 - c. Fibroblasts proliferate and migrate into the site of injury and lay down collagen fibers that form the scar.

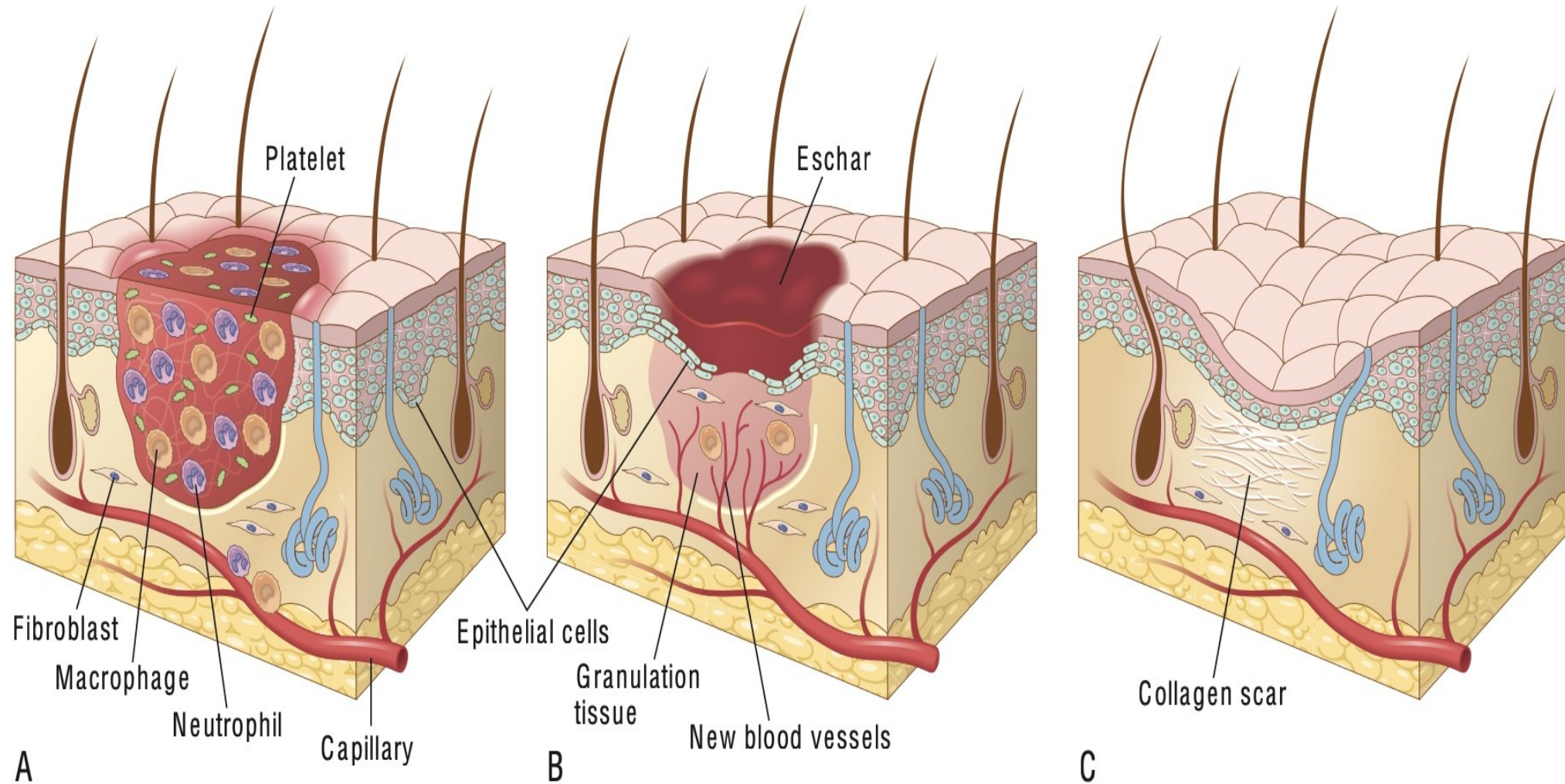


Figure 3.26 Steps in repair by scar formation: wound healing in the skin. (A) Inflammation. (B) Proliferation of epithelial cells; formation of granulation tissue by vessel growth and proliferating fibroblasts. (C) Remodeling to produce the fibrous scar.

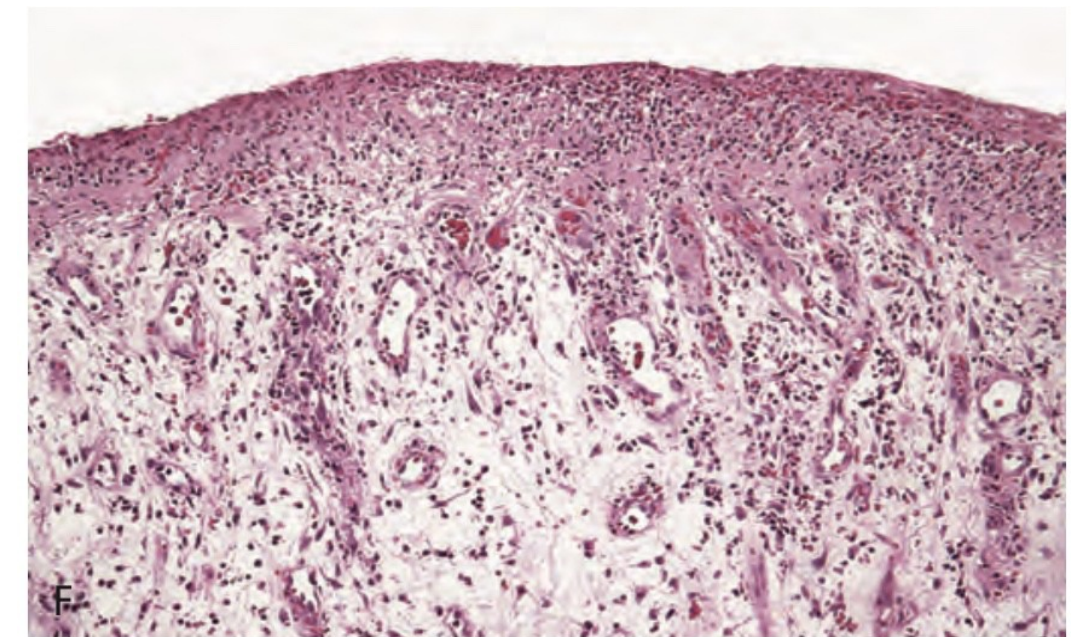
- **granulation tissue formation:**

- It is a highly vascularized loose connective tissue and a hallmark of early tissue repair it is peak at 5-7 days of healing and composed of

1. fibroblasts,
2. loose connective tissue
3. Blood Vessels
4. interspersed mononuclear leukocytes.

- The term granulation tissue derives from the **gross appearance:** pink, soft, granular, such as that seen beneath the scab of a skin wound.

- Its **histologic appearance** is characterized by proliferation of fibroblasts and new thin-walled, delicate capillaries (angiogenesis) in a loose ECM, often with admixed inflammatory cells, mainly macrophages.



Granulation tissue at 2 weeks is progressively replaced by deposition of collagen leading to formation of stable fibrous scar.

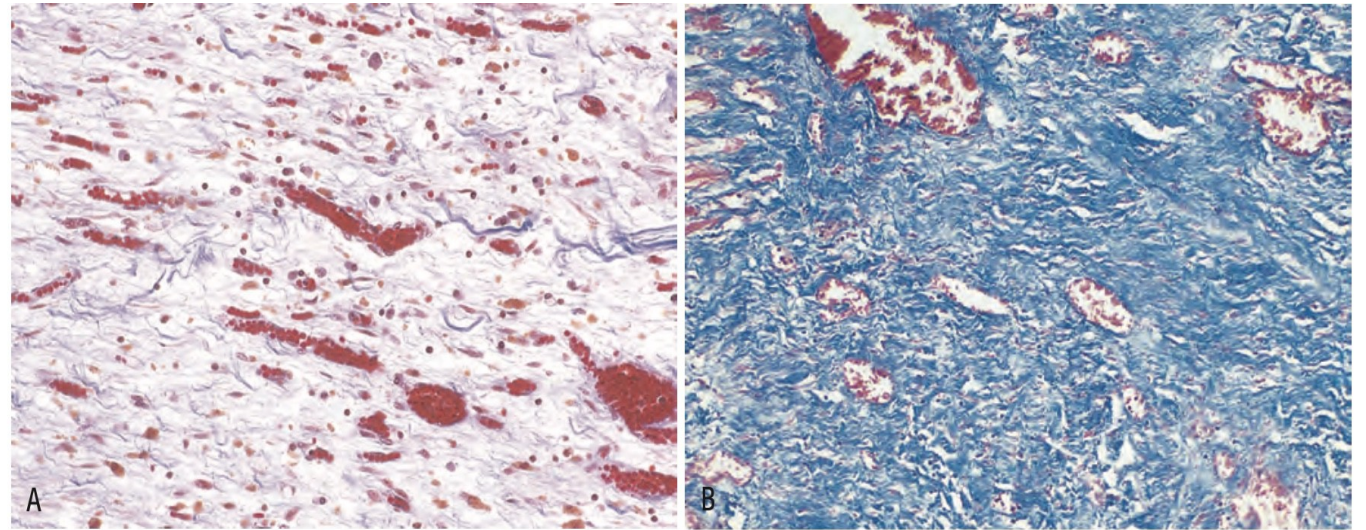


Figure 3.27 (A) Granulation tissue showing numerous blood vessels, edema, and a loose extracellular matrix containing occasional inflammatory cells. Collagen is stained blue by the trichrome stain; minimal mature collagen can be seen at this point. (B) Trichrome stain of mature scar, showing dense collagen, with only scattered vascular channels.

Angiogenesis

- Angiogenesis is the process of new blood vessel development from existing vessels. And consists of the following steps:

Vasodilation in response to nitric oxide and increased permeability induced by vascular endothelial growth factor (VEGF).

Separation of pericytes from the abluminal surface and breakdown of the basement membrane to allow formation of a vessel sprout.

Migration of endothelial cells toward the area of tissue injury.

Remodeling into capillary tubes.

Recruitment of periendothelial cells (pericytes for small capillaries and smooth muscle cells for larger vessels) to form the mature vessel.

- Anti VEGF medication like **AVASTIN** is used in treatment of cancer and diabetic retinopathy

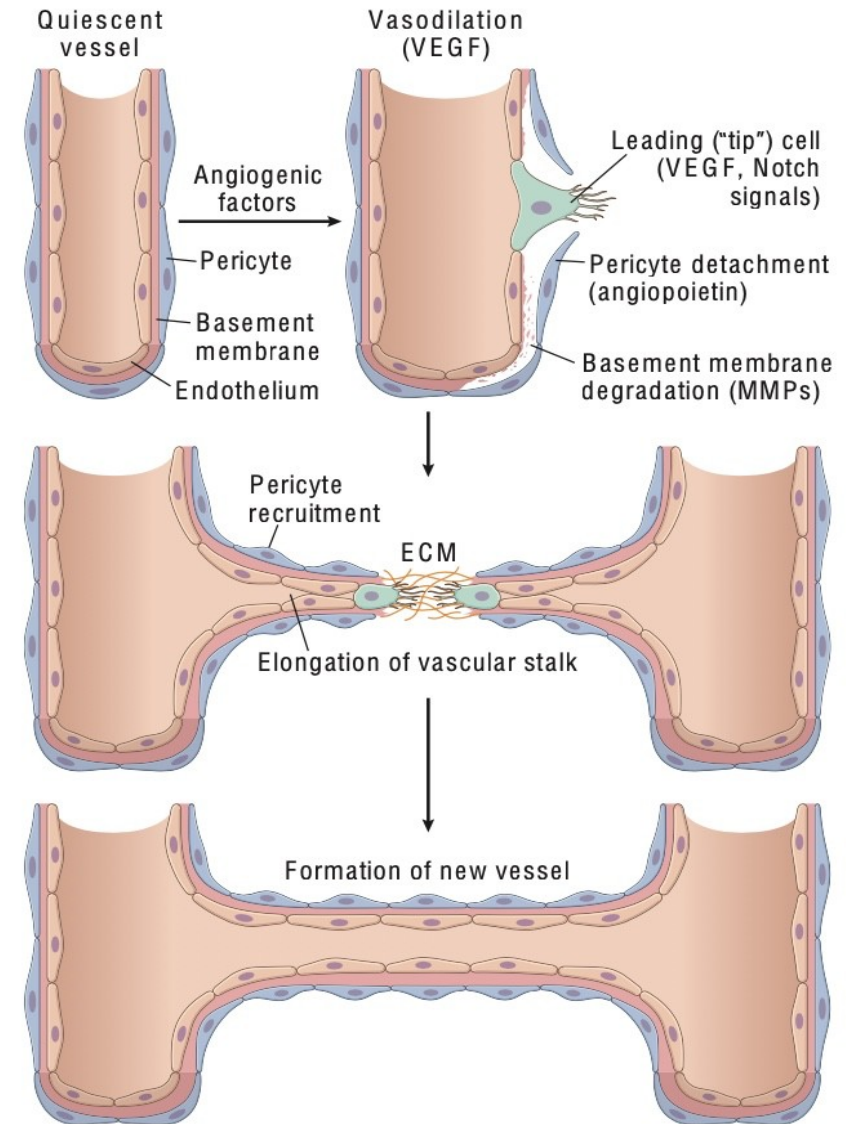


Figure 3.28 Angiogenesis. In tissue repair, angiogenesis occurs mainly by sprouting of new vessels. The steps in the process and the major signals involved are illustrated. The newly formed vessel joins up with other vessels (not shown) to form the new vascular bed. ECM, Extracellular matrix; MMPs, matrix metalloproteinases; VEGF, vascular endothelial growth factor.

Clinically Significant Types Of Repair—The Healing Of Skin Wounds

❖ Cutaneous wound healing is divided into three phases:

- *Inflammation (early and late);*
- *Granulation tissue formation and re-epithelialization;*
- *Wound contraction, ECM deposition, and remodeling.*

❖ Types of wound healing

1- Healing by primary/first intention:

- The edges of the wound are apposed to each other.
- Minimal loss of tissue.
- No infection.

2- Healing by secondary intention:

- The edges of the wound are widely separated and irregular.
- There is loss of a large amount of tissue.
- Infection is common in open wounds.

Healing by Primary Intension:

❖ This mode of healing occurs when the edges of the wound are closely opposed as in a surgical wound. The steps noted are:

- The wound gap is **immediately filled with blood and fibrin clot**.
- In the **first 24 hrs** the **neutrophils** invade the site.
- **24-48 hrs** the **epidermis begins to grow** from the most basal cells and form a thin layer.
- **3rd day:** by the end of the 3rd day the **granulation tissue** (it is a pink granular tissue that forms at the wound site and comprises of **new blood vessels, inflammatory cells and fibroblasts**) fills the wound and **macrophages** replace the neutrophils in order to clear up the inflammatory debris, thus the granulation tissue is a **vascularised connective tissue that heals the wound**.

Healing by Primary Intension:

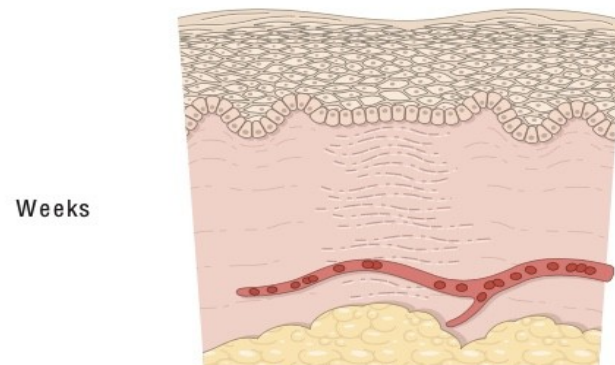
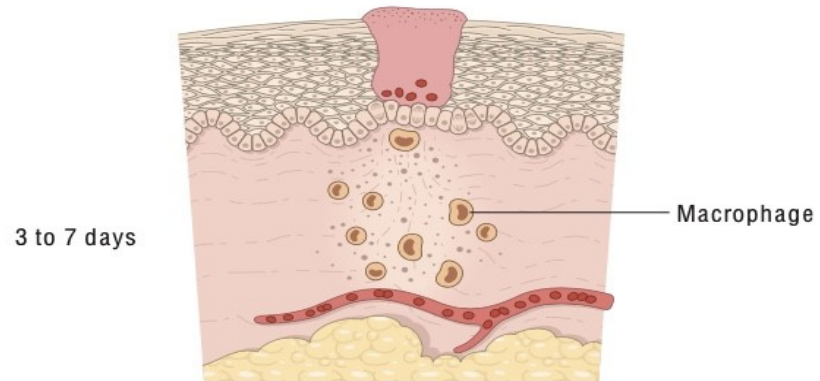
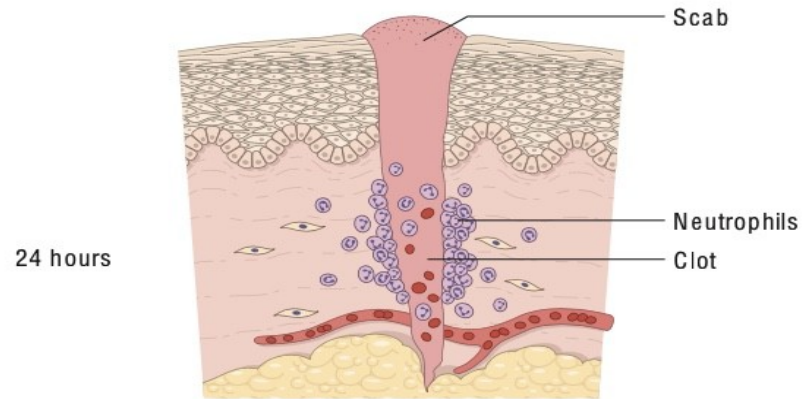
- By the **end of the 5th day** the **collagen fibers laid down** by the proliferating fibroblasts bridge the edges of the wound and **new vessels** are formed.
- By the end of **1st week -10th day** the **sutures are commonly removed** from skin wounds (the wound has **10% of tensile strength** of normal skin).
- By the **end of the 2nd week** the **inflammation and edema clears up** and a pale **fibrous scar** is formed.
- By **the end of the 1st month** the scar achieves **50% of its strength** and then on the strength is progressively increased until the **end of 3rd month** when it has **70-80 % of the normal tensile strength** of the skin.

Healing by Secondary Intention:

❖ It occurs when there is an extensive loss of tissue as in surface wounds that create large defects, reparative process is more complicated e.g. ulcers , large wounds, etc. It differs from healing by primary intention in the following features:

- *Inevitably large tissue defects generate a larger fibrin clot that fills the defect and more necrotic debris and exudate that must be removed, consequently the inflammatory reaction is more intense.*
- *The granulation tissue formed is large.*
- *The feature that most clearly differentiates primary from secondary healing is the phenomenon of wound contraction, which occur in large surface wounds due to presence of myofibroblasts, which have a smooth muscle like action.*
- *Substantial scar and thinning of the epidermis.*

HEALING BY FIRST INTENTION



HEALING BY SECOND INTENTION

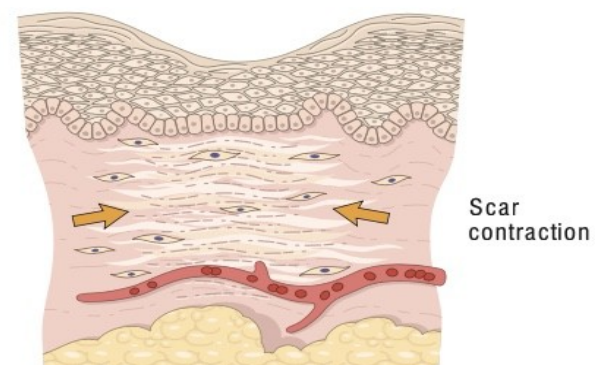
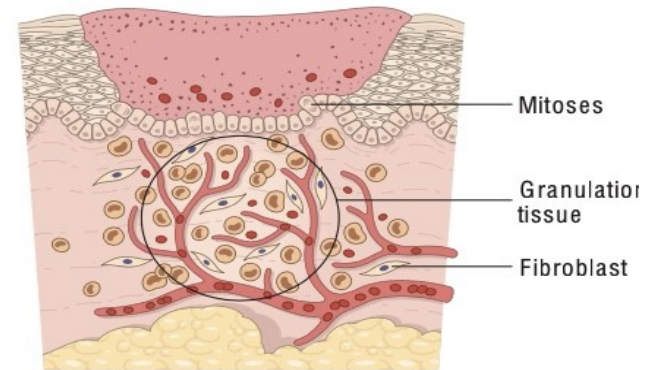
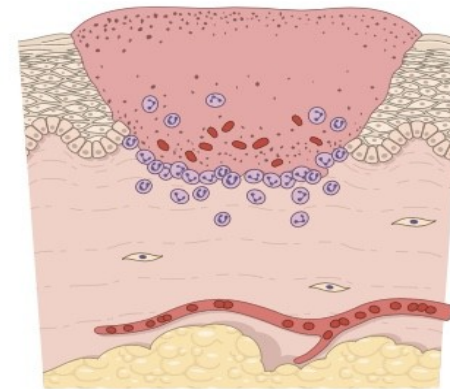


Figure 3.29 Steps in wound healing by first intention (left) and second intention (right). In the latter, note the large amount of granulation tissue and wound contraction.

Factors Influence Wound Healing

1. Nutritional status of the host (e.g., protein and vitamin C intake)
2. Metabolic status (diabetes mellitus delays healing)
3. Circulatory status or vascular adequacy
4. Size of wound
5. location (well-vascularized tissues heal faster)
6. clean wound heals faster than infected wounds
7. mechanical forces (e.g., motion)



Abnormality of healing

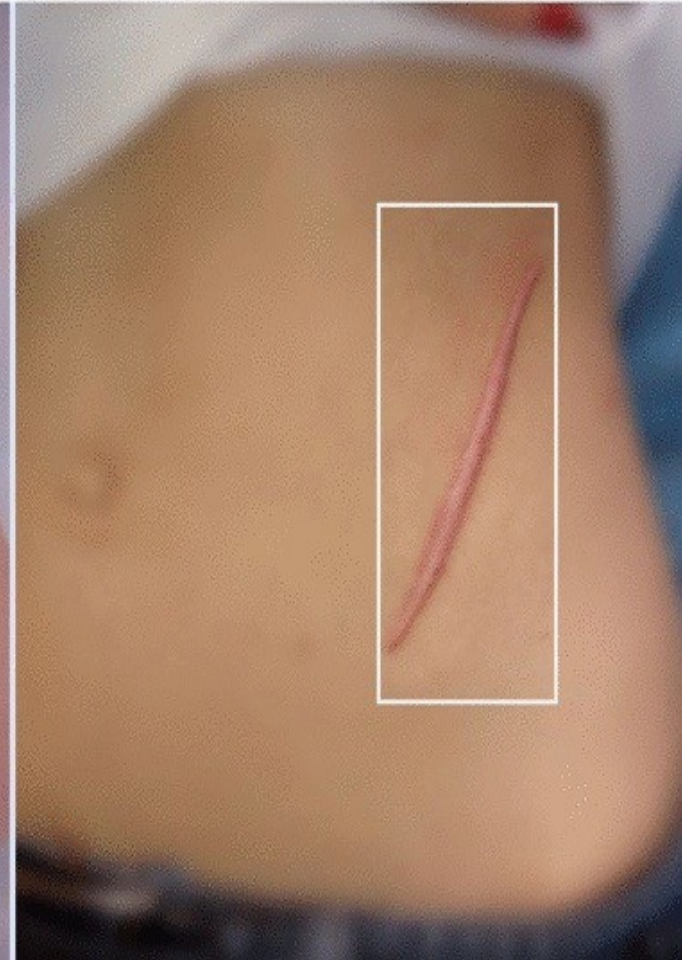
Excessive collagen-Scarring

- Excessive formation of the collagen (components of the repair proces) it is of two types:

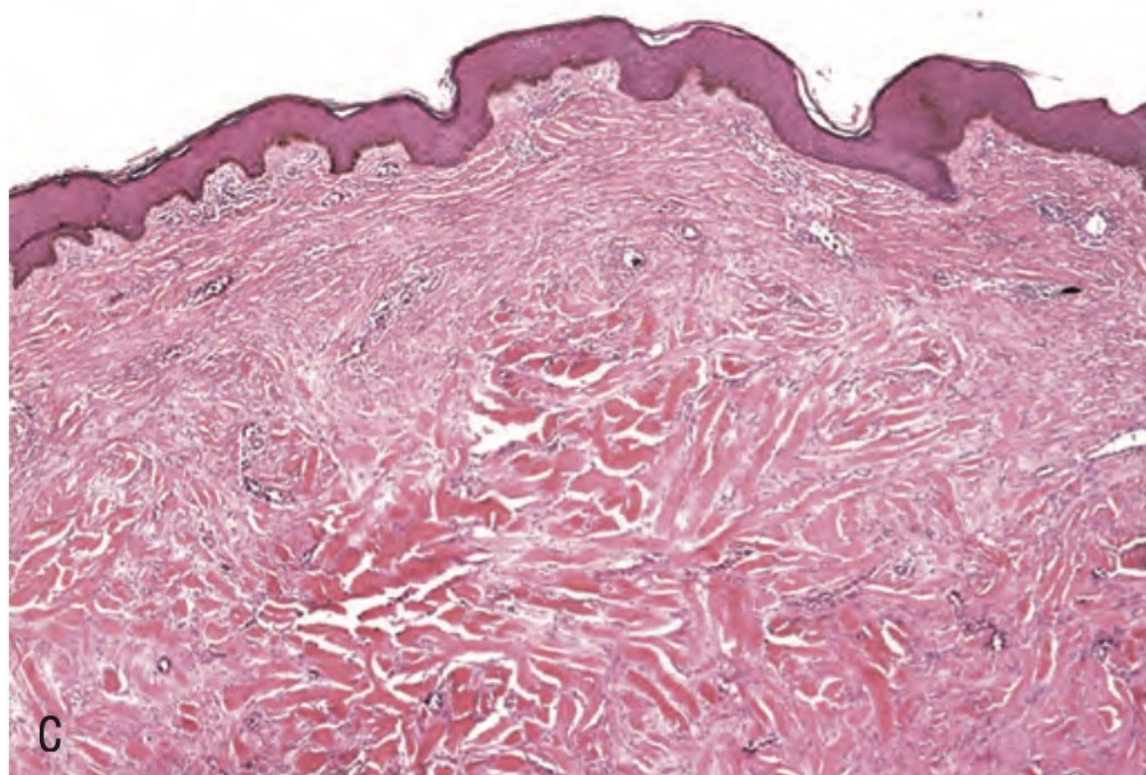
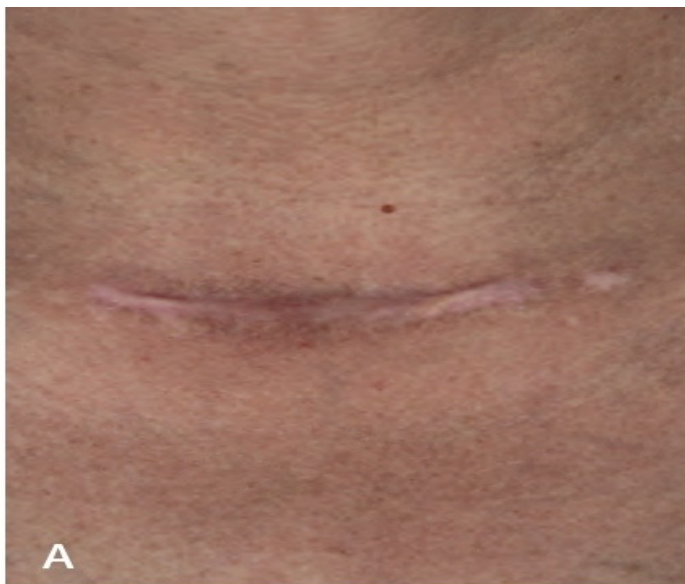
1. ***hypertrophic scar*** :The accumulation of excessive amounts of collagen may give rise to a raised scar. These often grow rapidly and contain abundant myofibroblasts, but they tend to regress over several months. develop after thermal or traumatic injury that involves the deep layers of the dermis.
2. ***keloid***: the scar tissue grows beyond the boundaries of the original wound and does not regress. It is more common in African American. In the ear lobe and on the chest.



Keloid



Hypertrophic



excessive granulation tissue

3. **proud flesh**: the formation of excessive amounts of granulation tissue, which protrudes above the level of the surrounding skin and blocks reepithelialization which must be removed to restore the epithelial tissue.

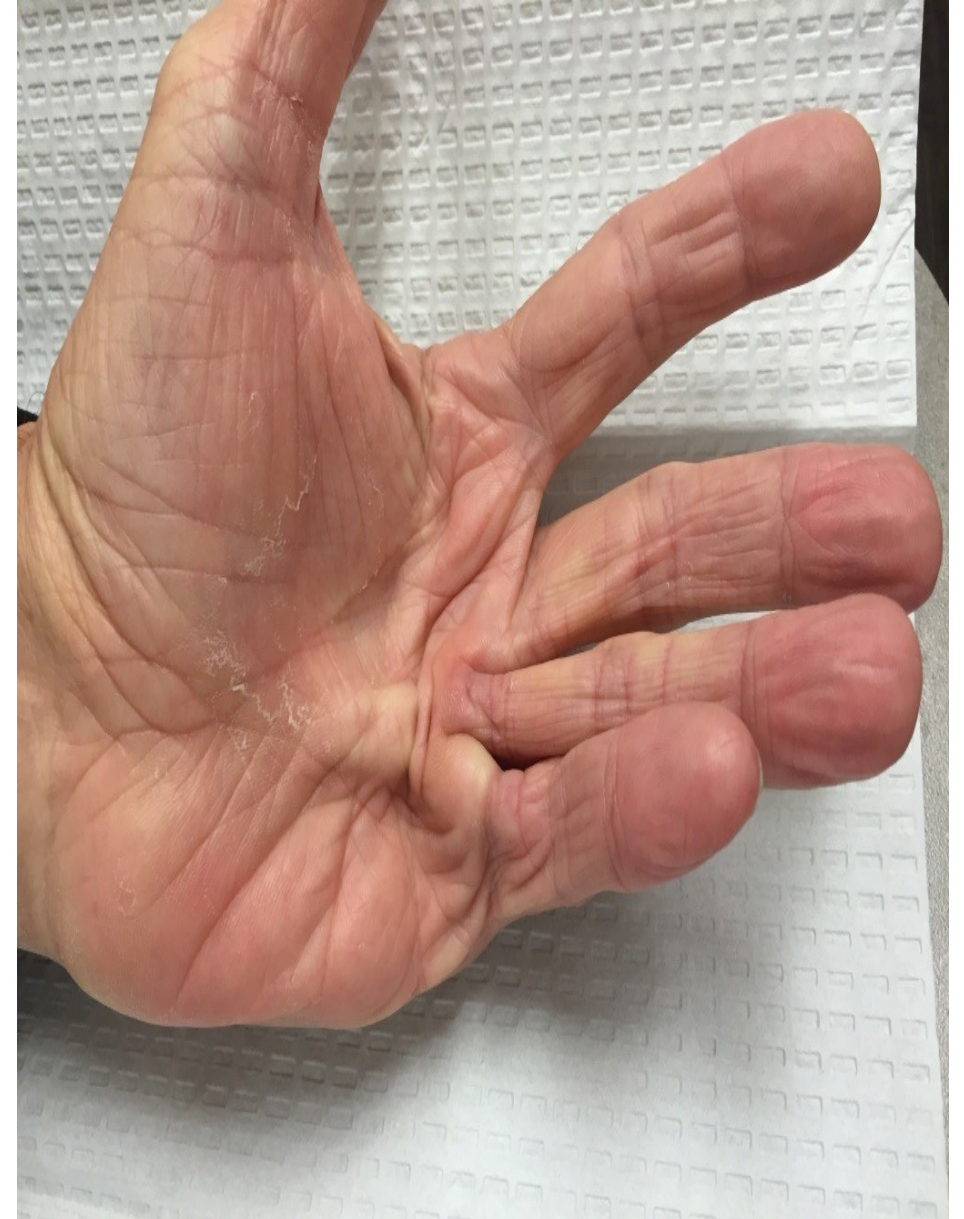
excessive fibroblast

3. **desmoid**: excessive proliferation of fibroblasts and other connective tissue elements after incisional scars or traumatic injuries it is rare condition characterized by recurrence after excision.



Contractures

- Although wound contraction is a normal part of healing, an exaggerated process is designated a contracture. It will cause wound deformity (e.g., producing hand claw deformities, or limit joint mobility). Contractures are particularly seen on the palms, the soles, and after serious burns.





Fracture

is defined as loss of bone integrity resulting from mechanical injury and/or diminished bone strength.

Fractures are the most common pathologic conditions affecting bone.

The following qualifiers describe fracture types and affect treatment:

- **Simple**: the overlying skin is intact
- **Compound**: the bone communicates with the skin surface
- **Comminuted**: the bone is fragmented
- **Displaced**: the ends of the bone at the fracture site are not aligned.

Stress: a slowly developing fracture that follows a period of increased physical activity in which the bone is subjected to repetitive loads

- **Greenstick**: extending only partially through the bone, common in infants when bones are soft
- **Pathologic**: involving bone weakened by an underlying disease process, such as a tumor.

Fracture

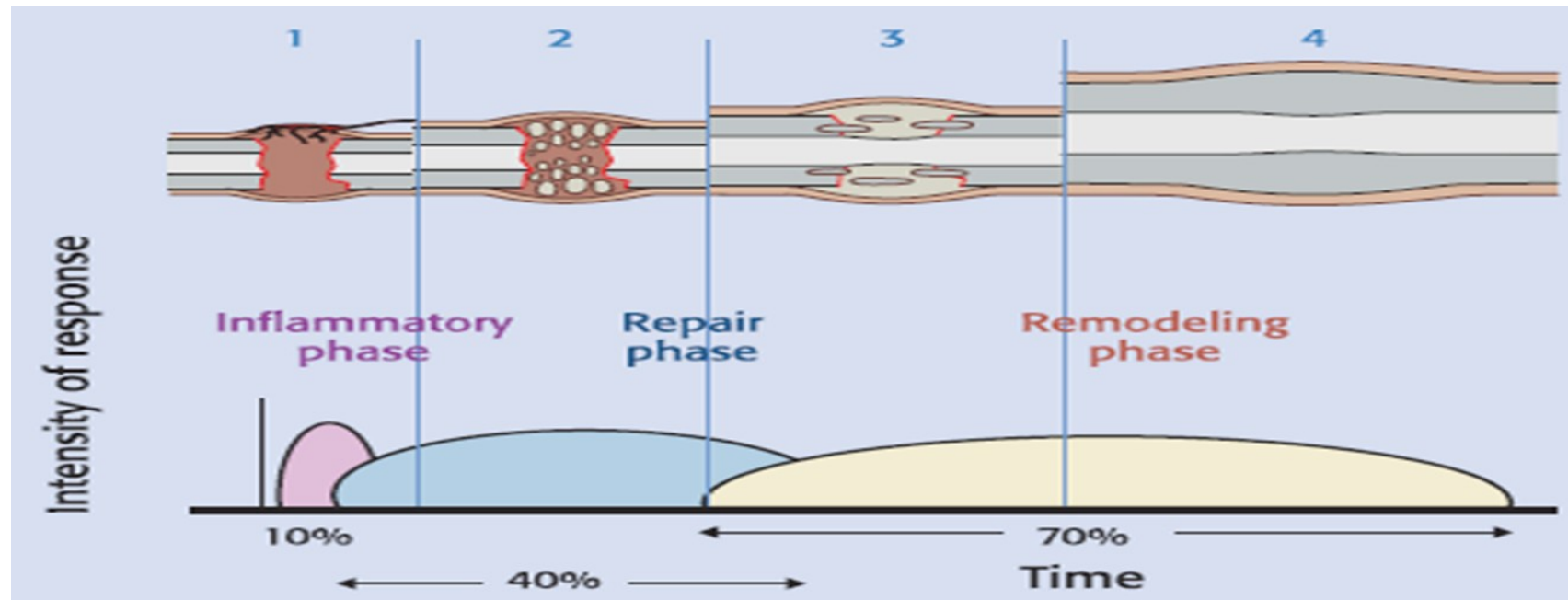
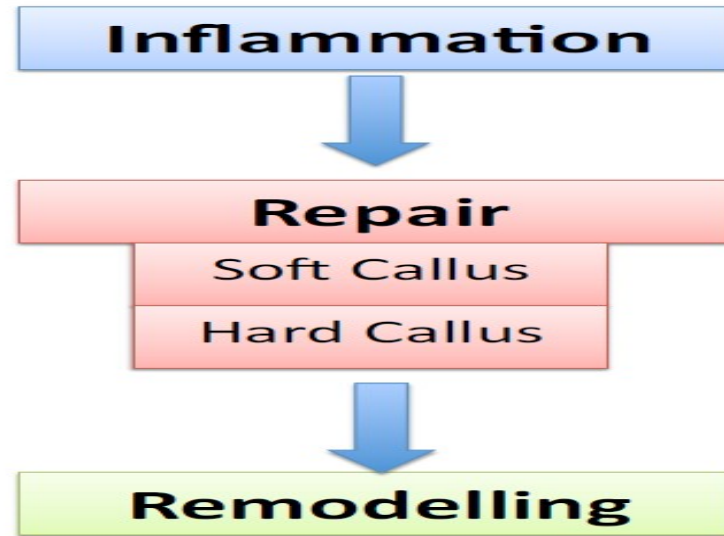
is defined as loss of bone integrity resulting from mechanical injury and/or diminished bone strength.
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fracture healing

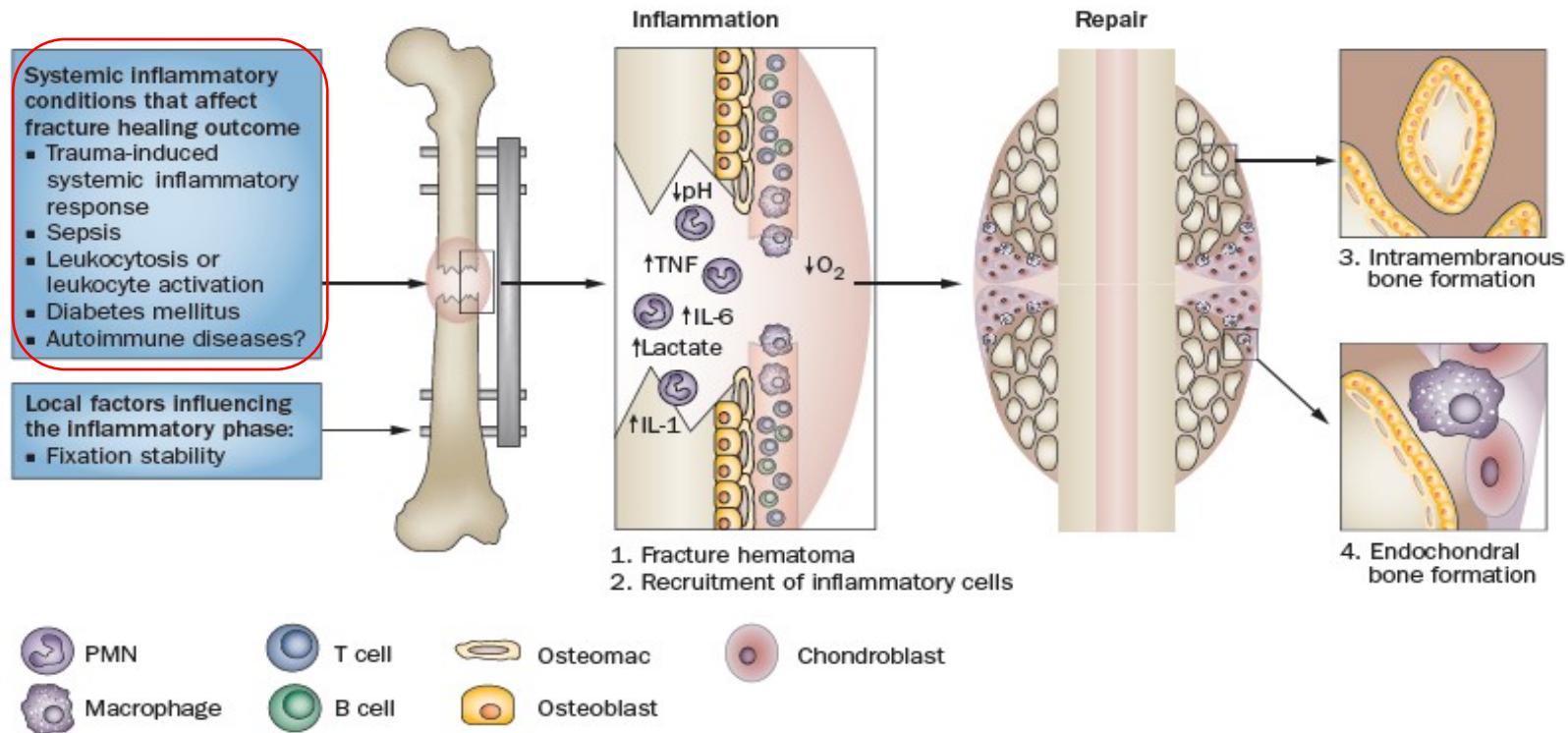
- Fracture healing shares many similarities with soft-tissue healing but its ability to be completed without the formation of a scar is unique.
- Fracture healing involves complex processes of cell and tissue proliferation and differentiation.
- Many players are involved, including growth factors, inflammatory cytokines, antioxidants, bone breakdown (osteoclast) and bone-building (osteoblast) cells, hormones, amino acids, and uncounted nutrients.

Healing of Fractures

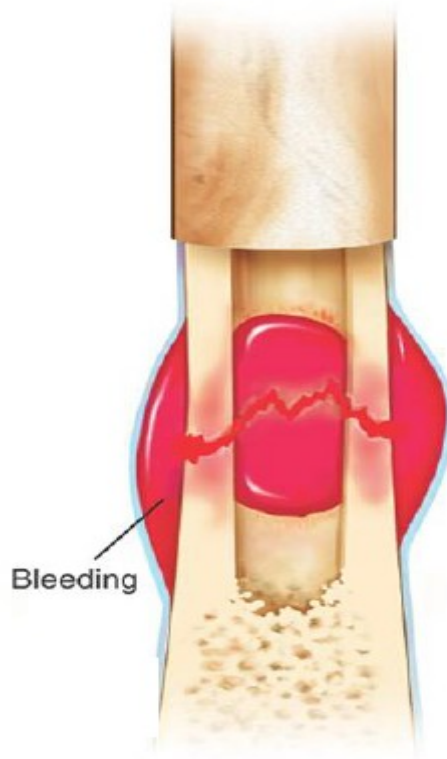
- ❑ Fracture repair involves regulated expression of a multitude of genes and can be separated into overlapping stages with particular molecular, biochemical, histologic, and biomechanical features.
- ❑ Bone healing, or fracture healing, is a proliferative physiological process in which the body facilitates the repair of a bone fracture. Fracture healing can be divided into three phases:
 - ❑ Inflammatory phase
 - ❑ Repair phase
 - ❑ Remodeling phase



- Schematic representation of inflammation and repair during fracture healing



- Inflammatory phase:
- The defect is initially filled with hematoma and there is intense inflammation



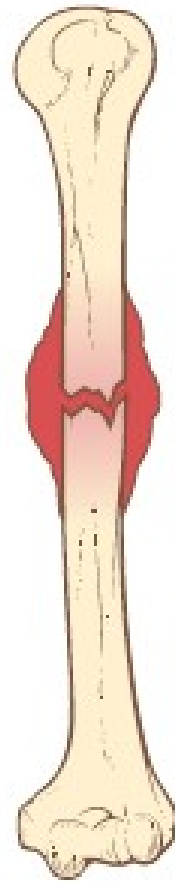
INFLAMMATION

- ✓ Hematoma (fibrin clot)
- ✓ Macrophages
- ✓ Inflammatory leukocytes
- ✓ Osteoprogenitor cells
- ✓ Hematopoietic cells secrete growth factors

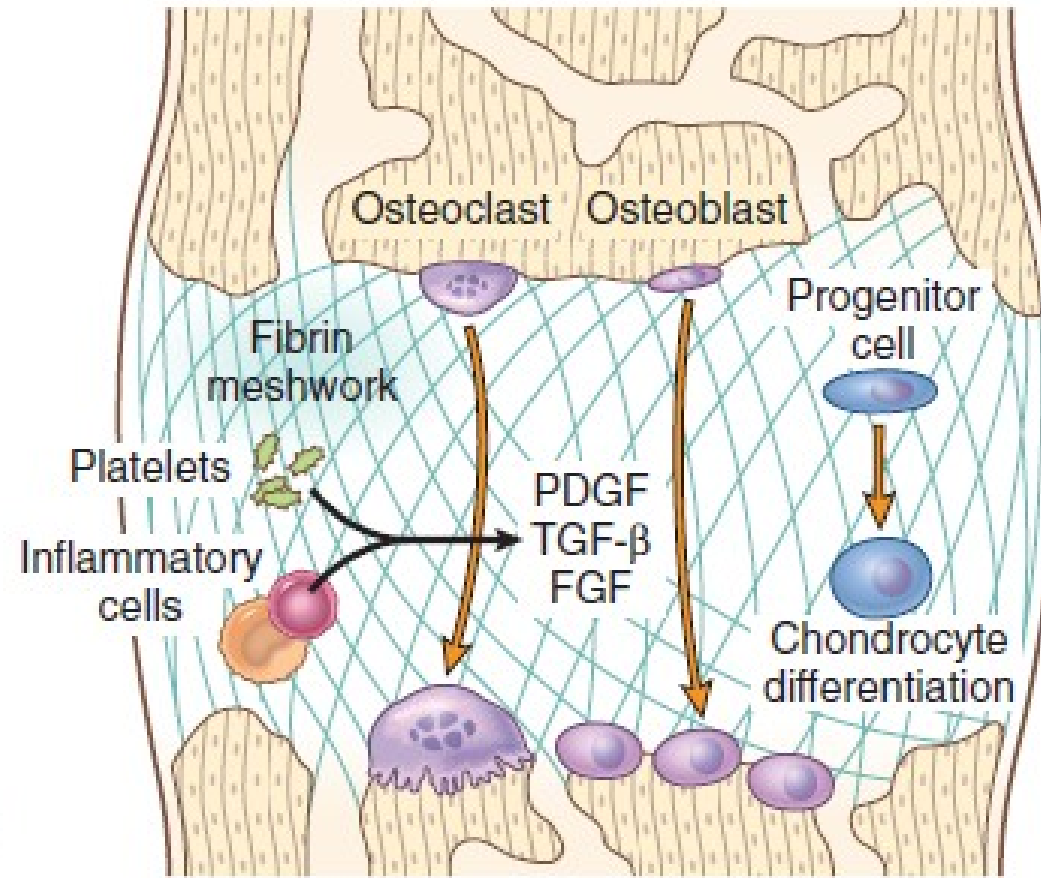
Granulation Tissue

Inflammatory

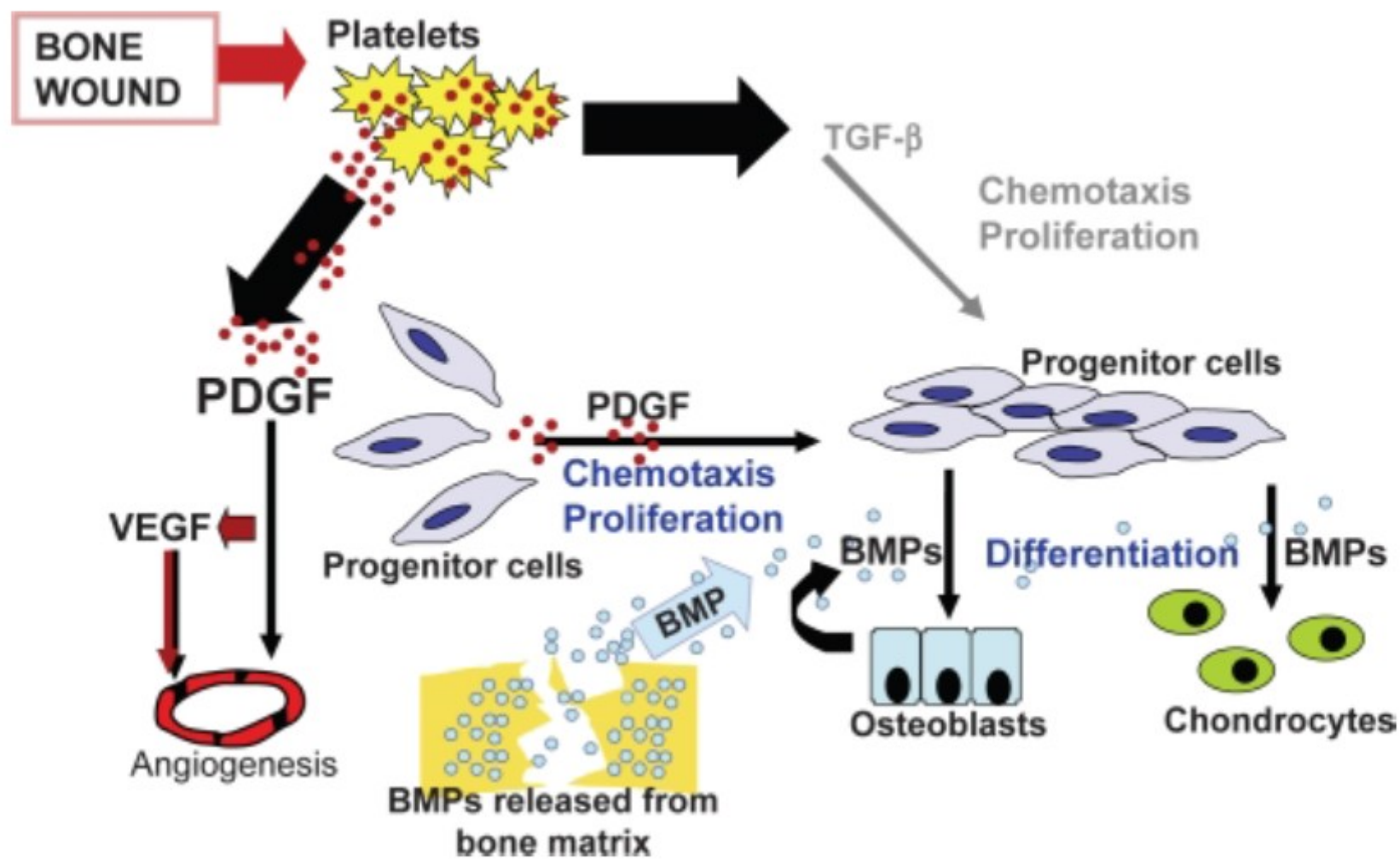
- Immediately after fracture, rupture of blood vessels results in a hematoma, which fills the fracture gap and surrounds the area of bone injury .
- The clotted blood provides a fibrin mesh, sealing off the fracture site and creating a scaffold for the influx of inflammatory cells and the ingrowth of fibroblasts and new capillaries
- degranulated platelets and migrating inflammatory cells release PDGF, TGF- β , FGF, and other factors that activate osteoprogenitor cells in the periosteum, medullary cavity, and surrounding soft tissues and stimulate osteoclastic and osteoblastic activity



**0-1 day
Organizing
hematoma**



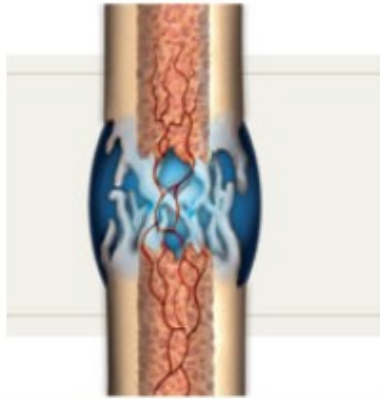
0-2 weeks - Soft callus



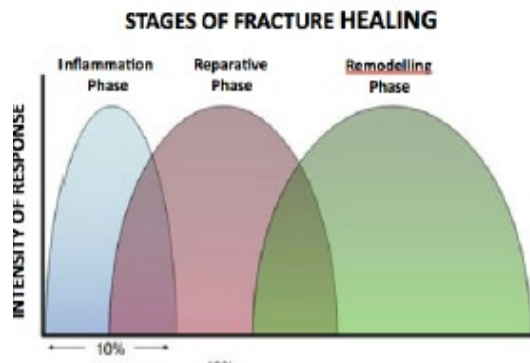
2. Repair phase:

This is quickly replaced by granulation tissue .

REPAIR



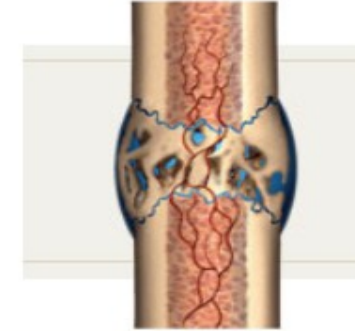
SOFT CALLUS



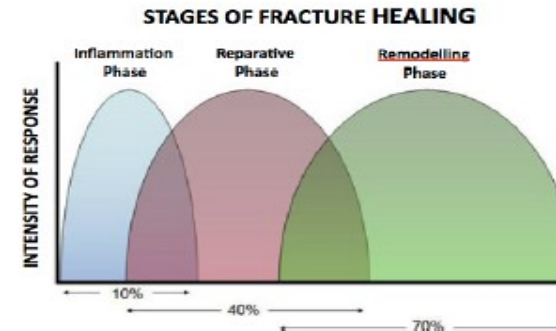
- ✓ Primary callus response (2 weeks)
- ✓ Inflammation triggers cell division and growth of new blood vessels (Angiogenesis)
- ✓ Chondrocytes secrete collagen and proteoglycans
- ✓ Creates fibrocartilage

REPAIR

- ✓ Enchondral ossification
- ✓ Bone formation
- ✓ Soft callus turns to hard callus (woven bone)



HARD CALLUS



- By the end of the first week,
a mass of predominantly uncalcified tissue— called soft callus or procallus—provides anchorage between the ends of the fractured bones.
- After approximately 2 weeks, the soft callus is transformed into a bony callus

The activated osteoprogenitor cells deposit woven bone.

In some cases, the activated mesenchymal cells in the soft tissues and bone surrounding the fracture line also differentiate into chondrocytes that make fibrocartilage and hyaline cartilage.

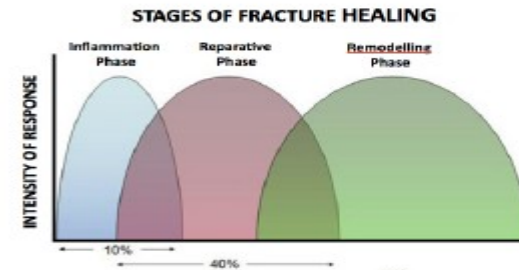
The newly formed cartilage along the fracture line undergoes endochondral ossification, forming a contiguous network of bone with newly deposited bone trabeculae in the medulla and beneath the periosteum.

- In this fashion the fractured ends are bridged
- Over the weeks a fibrocartilaginous callus is formed .
Mineralization leads to formation of a hard callus

3. Remodeling phase:
a hard callus becoming fusiform and slowly disappearing as Haversian remodeling progresses .

REMODELLING

- Begins during the middle of the repair phase
- Continues up to 7 years
- Allows the bone to assume its normal configuration
- The stresses on the bone is important (Wolff's Law)

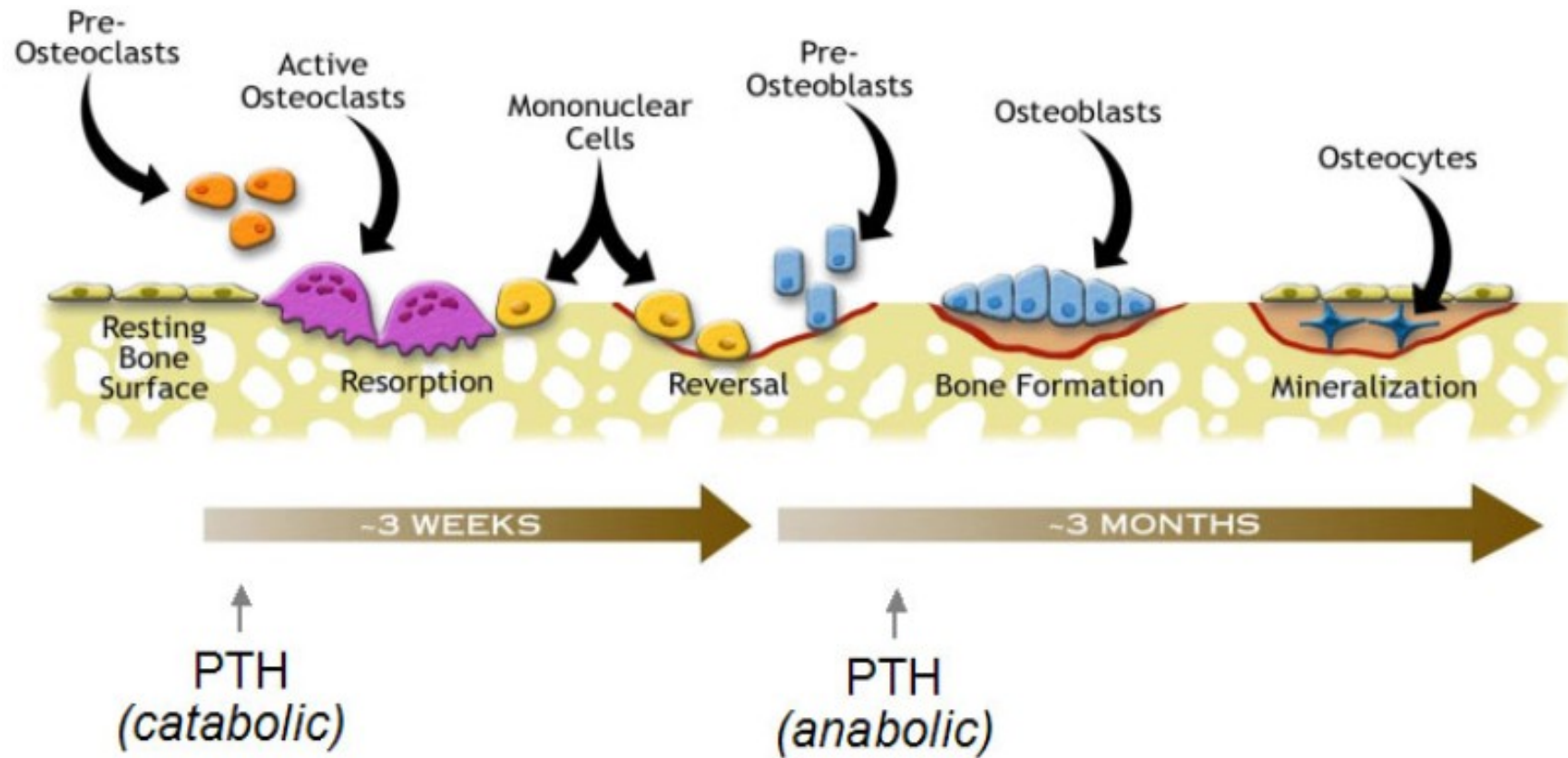


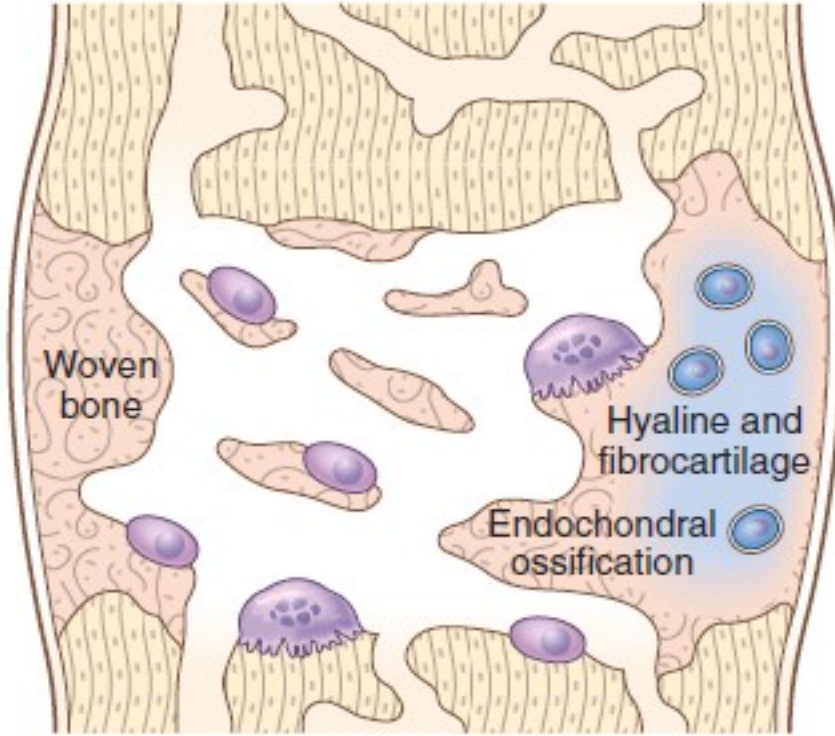
As the callus matures and is subjected to weight bearing forces, portions that are not physically stressed are resorbed.

This remodeling reduces the size of the callus until the shape and outline of the fractured bone are reestablished as lamellar bone.

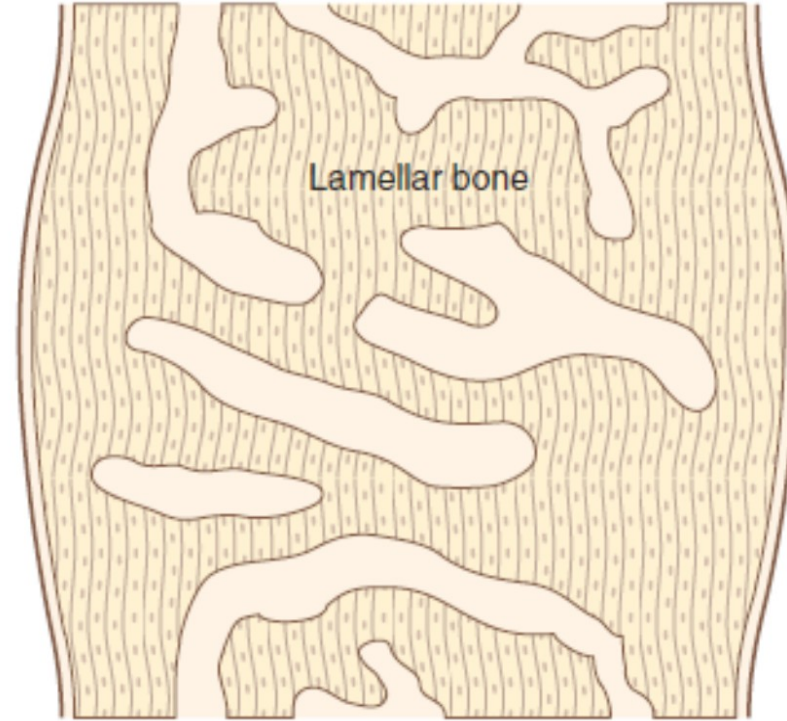
The healing process is complete with restoration of the medullary cavity.

Bone Remodelling Cycle



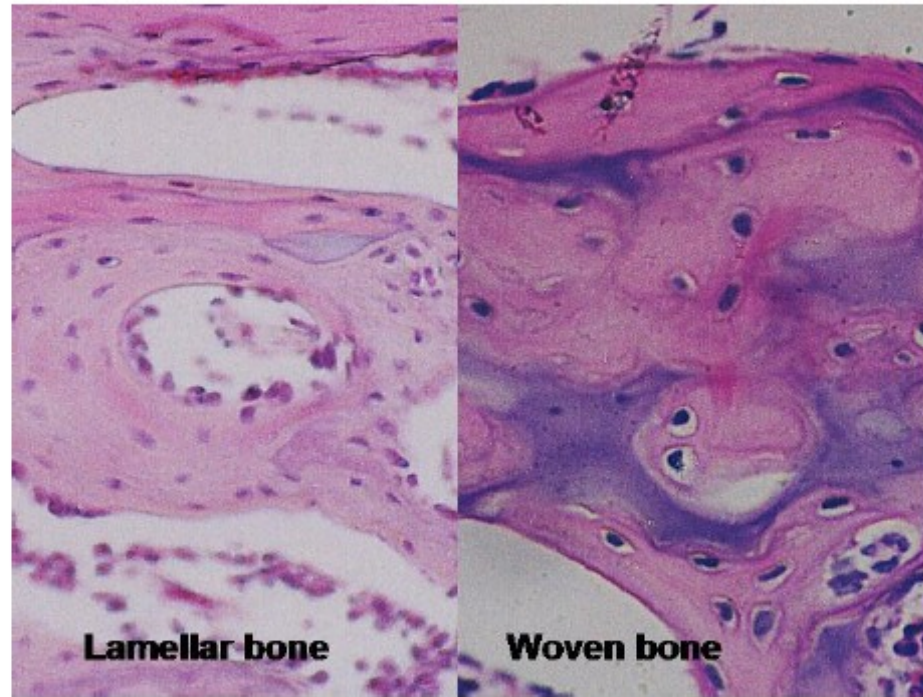


2-3 weeks - Bony callus



3 weeks-months - Bony callus

REMODELLING



- Woven bone replaces with lamellar bone

Factors that affect fracture healing

- Many risk factors for impaired fracture healing exist:
 - Type of injury (fracture geometry, degree of open injury, mechanism of injury).
 - Fracture treatment (type of fixation, size of fracture gaps);
 - Gender, age.
 - Comorbidities (diabetes mellitus, malnutrition, peripheral vascular disease, hypothyroidism, polytrauma).
 - Medications (NSAIDs, corticosteroids, antibiotics, anticoagulants); smoking; and alcohol consumption.
 - Infection of the fracture site, especially common in open fractures, is a serious obstacle to healing
 - Inadequate immobilization permits movement of the callus and prevents its normal maturation, resulting in delayed union or nonunion. If a nonunion persists, the malformed callus undergoes cystic degeneration, and the luminal surface may become lined by synovial-like cells, creating a false joint or pseudoarthrosis.